

On the Security of Linear Secret Sharing with General Noisy Side-Channel Leakage

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Abstract. Secret sharing is a foundational cryptographic primitive for sharing keys in distributed systems. In a classical (n, t) -threshold setting, it involves a dealer who has a secret, a set of n users to whom shares of the secret are sent, and a threshold t which is the minimum number of shares required to recover the secret. These schemes offer an *all-or-nothing* security approach where less than t shares reveal no information about the secret. But these guarantees are threatened by side-channel attacks which can leak partial information from each share. Initiated by Benhamouda et al. (Crypto’18), the security of linear secret sharing schemes has been studied for bounded leakage attack models, which assume that the adversary can leak bounded functions of each share. However, this model does not translate into real-world attacks, as physical side-channels are inherently noisy. The δ -noisy channel model, proposed by Prouff and Rivain (Eurocrypt’13), is a general leakage framework which captures the noisy behavior of side-channels. In this work, we study the security of linear secret sharing schemes with δ -noisy leakage, and show bounds on the mutual information (MI) and statistical bias (Δ^{TV}) security metrics. Our results are based on the Fourier analytical framework, first used by Benhamouda et al. (Crypto’18), adapted to the δ -noisy leakage model. To give security bounds, we introduce a security parameter $\eta \leq \min\{1, 2\delta\}$, determined by the Fourier linear biases of the posterior distributions of the leaked secret shares. Then, the Poisson summation formula enables us to bound the ratio between the observed leakage for some given secret, and leakage under independence as $(1 \pm \eta^t)$. This is then used to show a) $(n, t \geq \tau(n+1))$ -threshold schemes over $\mathbb{F}_q$ have at most $\mathcal{O}(q^{-t(\gamma+1-1/\tau)})$ leakage, given $\eta \leq q^{-\gamma}$; and consequently b) for (n, n) -threshold schemes the guessing advantage is at most $(q-1) \cdot \eta^n \leq (q-1) \cdot (2\delta)^n$. This work can be viewed as a next step towards closing the gap between theory and practice in leakage resilient cryptography.

Keywords: Secret sharing · Side-channel leakage · Information-theoretic cryptography · Leakage-resilient cryptography.

1 Introduction

The design and analysis of cryptographic primitives often assume cryptosystems as impervious black-boxes. However, in real-world applications, adversaries can

bypass theoretical security guarantees by exploiting the implementation and physical environment of such systems. Side-channel attacks constitute a class of cryptanalytic techniques that leverage the operational behavior of hardware systems such as power consumption, electromagnetic emanations, etc., to reveal sensitive information. Such adversarial attacks on fundamental cryptographic building blocks undermine the security guarantees of all computer systems that incorporate them. Consequently, protocols that attempt to provide provable security guarantees against side-channel attacks have attracted significant attention in the cryptographic community (survey [26]).

Secret sharing, introduced by Shamir [47] and Blakley [8], is a fundamental cryptographic primitive central to security in distributed systems. It protects against collusion by distributing *shares* of a secret among parties/users such that only specific subsets of users, called access structures, can reconstruct the secret by combining their shares. In the classical (n, t) -threshold setting, shares of the secret are distributed among n users such that all subsets of size at least t constitute the access structures. Despite their adoption in several real-world systems (survey [5]), the impact of threshold, noise, and scalability on the security of such schemes is not fully understood. In this work, we study the security guarantees of (n, t) -threshold secret sharing schemes constructed using linear codes, where a *noisy* version of each secret share is leaked to an adversary.

To characterize this leakage, we consider the δ -noisy side-channel leakage model proposed by Prouff and Rivain in Eurocrypt’13 [46]. This model assumes that the adversary obtains a noisy version of every secret share, leaked through independent side-channels, with some leakage parameter δ . Since this model closely captures real-world side-channel attacks, it has become the standard leakage framework for side-channel security analysis. Masking schemes, which usually rely upon (n, n) -threshold schemes, are used extensively as a countermeasure to protect cryptosystems against side-channel leakage. Consequently, several works have investigated the security guarantees of (n, n) -threshold schemes under δ -noisy leakage [15, 17, 24]. In this work, we generalize the analysis to general thresholds, and show bounds on the security of (n, t) -threshold secret sharing schemes with δ -noisy leakage (Theorem 4).

Research in leakage-resilient (n, t) -threshold schemes has largely focused on models where the adversary can leak arbitrary bounded functions of the secret shares [6, 7, 27, 28, 31, 32, 33, 35, 41]. Such models characterize *worst-case* discrete leakage and do not directly capture real-world side-channel attacks [48, 49]. This results in conservative security guarantees, and that too with severe constraints on the underlying field $\mathbb{F}_q$. In practice, side-channels are inherently noisy and techniques exist to amplify this noise [36]. Therefore, this work should be seen as a step in studying the security of cryptographic primitives with more realistic leakage models. Furthermore, existing works usually prove formal security guarantees only for secret sharing schemes with thresholds $t > n/2$, even though schemes with $t \leq n/2$ are often used in practice. Therefore, understanding the security of schemes with general thresholds, and implemented over arbitrary finite fields, for noisy adversarial models is necessary.

1.1 Related Prior Work

In the context of leakage resilience of (n, t) -threshold secret sharing, most research has focused on bounded leakage models [6, 7, 27, 28, 31, 32, 33, 35, 41]. Guruswami and Wootters’ reconstruction algorithm [21] showed that even a single *adversarial* bit leaked from each secret share compromises the security of Shamir’s secret sharing scheme over fields of characteristic 2. This led Benhamouda et al. [6, 7] to investigate the leakage resilience of linear secret sharing schemes for other finite fields. They prove that the (n, t) -Shamir’s scheme is leakage-resilient against one bit leakages, when the underlying field is of a large prime order, and $t \geq 0.92n$. This ratio has been progressively improved to 0.668 in a series of works by different authors [7, 27, 28, 30, 33]. For the noisy probing model, Adams et al. [1] proved that for additive schemes over a prime field $\mathbb{F}$, n must be at least $\omega(\log \lambda / \log \log \lambda)$ where $\lambda = \lceil \log_2 |\mathbb{F}| \rceil$ to ensure security against single physical-bit leakage. However, a general drawback of such models is the restrictions required on the underlying field $\mathbb{F}_q$. For example, state-of-the-art results by Kasser [27] require a prime field $\mathbb{F}$ with size $|\mathbb{F}| = \mathcal{O}(2^n)$. In another series of works [34, 35, 41], it was shown that Shamir’s secret sharing scheme constructed using random evaluation points is almost always secure against physical probing attacks over all finite fields irrespective of size or characteristic. This result seems to suggest that an (n, t) -threshold scheme with *random/noisy* leakage, might also be secure over all fields. In this work, we answer this affirmatively.

Noisy Leakage Models. With applications to masking schemes, the security of (n, n) -threshold schemes has been studied extensively against diverse noisy leakage models [2, 3, 4, 9, 12, 14, 15, 17, 18, 19, 20, 24, 25, 29, 39, 43]. For (n, n) -threshold schemes with δ -noisy leakage, for arbitrary fields, the best known bounds are due to Dziembowski, Faust, and Skórski (TCC’16) [17]. They show that the security of (n, n) -threshold schemes improves exponentially with the number of shares. This is done by first reducing the security analysis to uniform secrets and consequently independent secret shares, and then using the XOR lemma. However, for general thresholds, the secret shares are not independent even if the secret is uniform, making it difficult to extend these proof techniques. In this work, we instead take a direct approach towards proving security guarantees by invoking the Poisson summation formula from discrete Fourier analysis.

Information Leakage Measures. In cryptanalysis literature, mutual information (MI) and statistical bias (Δ^{TV}), are widely regarded as the fundamental metrics for quantifying leakage in side-channel attacks [45, 48]. These metrics provide *operationally meaningful* insights, letting hardware engineers translate system design into measurable security guarantees. However, since these metrics are often difficult to bound directly, several ad hoc surrogate metrics have been proposed to characterize security in leakage-resilient cryptography. In this work, besides showing bounds for MI and Δ^{TV} , we additionally consider the Complementary Doeblin Coefficient (Δ^{CDC}) metric [3], and the Average Relative Error (Δ^{ARE}) metric [45]. Bounding the Δ^{ARE} metric to characterize the overall leakage is particularly useful, since $\Delta^{\text{ARE}} \geq \{2 \cdot \Delta^{\text{TV}}, \frac{\text{MI}}{H(X)}, \Delta^{\text{CDC}}\}$ [3, 45].

1.2 Our Contributions

The main result of our work provides formal security guarantees for (n, t) -threshold linear secret sharing schemes with δ -noisy leakage. Consequently, we also improve the guarantees for masked encodings for a small leakage parameter δ . To explain our results, we begin by introducing some mathematical notation.

Linear Code-Based Secret Sharing. Let $\mathcal{M} \subseteq \mathbb{F}_q^{n+1}$ be a linear code over a finite field $\mathbb{F}_q$. The secret sharing scheme for n parties based on $\mathcal{M}$ is constructed as follows. Let the secret be $s \in \mathbb{F}_q$. Sample a random codeword $(s, u_1, \dots, u_n) \in \mathcal{M}$, and assign the secret share u_i to the i^{th} user.

δ -Noisy Leakage Model. This model assumes that each share is leaked to an adversary through independent δ -noisy channels with parameters δ_i . Formally, for a leakage metric Δ^{sha} , the adversary obtains leakage $\mathbf{L} = (L_1, \dots, L_n)$ such that $\Delta^{\text{sha}}(U_i; L_i) \leq \delta_i$ for all i . Our goal is to bound the overall leakage $\Delta^{\text{sec}}(S; \mathbf{L})$ for a given security metric Δ^{sec} .

Leakage Metrics. With a constant factor reduction, the results in this section are equivalent for the following metrics: $\Delta^{\text{sha}} \in \{\Delta^{\text{TV}}, \Delta^{\text{CDC}}, \text{MI}^{1/2}, \Delta^{\text{ARE}}\}$, and $\Delta^{\text{sec}} \in \{\Delta^{\text{TV}}, \Delta^{\text{CDC}}, \text{MI}, \Delta^{\text{ARE}}\}$ where $\Delta^{\text{TV}}(A; B)$ denotes the TV (total variation) distance between distributions A and $A \mid B$, MI is the mutual information metric, Δ^{CDC} is the complementary Doeblin coefficient [3], and Δ^{ARE} is average relative error metric [45].

Security Parameter $\eta_{\mathcal{M}}$. For an (n, t) -threshold scheme under δ -noisy leakage, we define the security parameter $\eta_{\mathcal{M}}$ via the Fourier biases of the posterior distributions $\mathbb{P}(U_i | L_i = \ell_i)$ of the leakage channels (see Definition 16). For this discussion, it suffices to note that the security parameter satisfies $\eta_{\mathcal{M}} \leq \min \left\{ 1, 2 \cdot \max_{\mathcal{J} \in \binom{[n]}{t}} \left(\prod_{j \in \mathcal{J}} \delta_j \right)^{1/t} \right\}$, where $\binom{[n]}{t}$ denotes the collection of all t -element subsets of $\{1, \dots, n\}$.

Security of (n, t) -Threshold Schemes. For an (n, t) -threshold scheme, define the normalized threshold as $\tau = t/(n+1)$. Then, for some given noise parameter $\eta_{\mathcal{M}}$, the main result of this work (Theorem 4) can be informally stated as.

Theorem 1 (Informal). *The information leaked about the secret S , is $\Delta^{\text{sec}}(S; \mathbf{L}) = \mathcal{O}(q^{(n+1-t)\eta_{\mathcal{M}}^t})$. For $\tau > 1/(1 - \log_q(\eta_{\mathcal{M}}))$, the leaked information is $\mathcal{O}(q^{-\Omega(t)})$.*

Theorem 1 shows that for a sufficiently small leakage parameter δ , linear secret sharing schemes are leakage resilient for arbitrarily small normalized thresholds τ . Countermeasures such as hiding, and noise amplification can be used to make the noise parameter $\eta_{\mathcal{M}}$ appropriately smaller.

Corollary 1 (Informal). *For normalized threshold $\tau = t/(n+1)$, if $\eta_{\mathcal{M}} \leq q^{-\gamma}$, where $\gamma > \frac{1}{\tau} - 1$, then the leakage from the secret is $\Delta^{\text{sec}}(S; \mathbf{L}) = \mathcal{O}(q^{\frac{1}{\tau} - 1 - \gamma})$.*

For the bounded leakage model, in a series of works [6, 7, 27, 28, 33, 35], Shamir's secret sharing scheme has been shown to be resilient only for $t \geq 0.668n$. However, as noted earlier, bounded leakage models do not translate into real-world

side-channel attacks. In Section 4, we formally recall the δ -noisy leakage model, and discuss its connection to other works.

Security of (n, n) -Threshold Schemes. Motivated by applications in masked encodings, the security of (n, n) -threshold schemes with δ -noisy leakage has been studied extensively in the literature. For arbitrary finite fields, the best known bounds are due to Dziembowski, Faust, and Skórski [17]. For a prime group $\mathbb{Z}_p$, they show that the minimum required number of shares to guarantee security is $n > 2 \ln p \cdot (1 - 1/p - \delta)^{-4}$. For small leakage parameters δ however, this bound is too large. As a consequence of Theorem 1, we improve this to $n > -\ln p / \ln 2\delta$.

Corollary 2 (Informal). *For (n, n) -threshold schemes, the information leaked about the secret is $\Delta^{\text{sec}}(S; \mathbf{L}) = \mathcal{O}(q^{1/n} \cdot \eta_{\mathcal{M}})^n$ and $\Delta^{\text{sec}}(S; \mathbf{L}) = \mathcal{O}(2^n \cdot q \cdot \prod_{i=1}^n \delta_i)$.*

For some binary extension field, $\mathbb{F}_{2^\lambda}$, Corollary 2 implies that, even a small number of shares $n = c\lambda$, give us exponentially strong security guarantees $\mathcal{O}(2^{1+1/c} \cdot \delta)^{c\lambda}$. This bound allows us to design protocols with concrete security guarantees for practical implementations of (n, n) -threshold schemes; such as for AES $q = 2^8$, Kyber $q \approx 2^{12}$ etc., and $n = 2, 3, 4$.

Impact. Our work can be viewed as a next step towards closing the gap between theory and practice in leakage resilient cryptography. For (n, n) -threshold schemes over arbitrary fields, we improve the security guarantees when the leakage parameter δ is small. This makes the theory of side-channel analysis practically relevant in designing provably secure implementations of secret sharing schemes. For (n, t) -threshold schemes, to the best of our knowledge, this is the first attempt towards understanding their security with practically relevant noisy leakage models. We present and discuss our results in Section 5.

1.3 Technical Overview

The extant analysis of leakage-resilient (n, t) -threshold secret sharing has been done using a Fourier analytic framework introduced by Benhamouda *et al.* in Crypto’18 [6]. Methods from Fourier analysis have been used extensively in additive combinatorics (see Tao and Vu [50]). For the bounded leakage model, several works [6, 7, 33] use a point-by-point analysis to bound the total variation distance based measures. We adapt these Fourier analytical techniques to noisy leakage frameworks, to show security guarantees for (n, t) -threshold schemes.

Problem Setup. Formally, let $x \leftarrow X$ denote a random variable X , and its realization x . For a linear (n, t) -threshold scheme over $\mathbb{F}_q$, let $s \leftarrow S$ be the secret, and let $\mathbf{L} = (L_1, \dots, L_n)$ be the vector of noisy secret shares leaked to the adversary. Under the δ -noisy leakage model, shares leak information to the adversary through independent side-channels characterized by leakage parameters δ_i ’s. For some leakage metric, Δ^{sha} , this can be written as $\Delta^{\text{sha}}(U_i, L_i) \leq \delta_i$, where $\mathbf{u} \leftarrow \mathbf{U}$ is the vector of secret shares. For some security metric Δ^{sec} , our goal is to bound $\Delta^{\text{sec}}(S; \mathbf{L})$. In this analysis, we will bound the average relative error leakage metric, Δ^{ARE} , first studied in the context of (n, n) -threshold

schemes by Prest *et al.* in Crypto'19 [45]. The authors show that bounding Δ^{ARE} implies security (with a constant reduction factor) for the mutual information (MI) and total variation (TV) based metrics (Section 3, Lemmas 2 and 3).

Approach. First, we express the security metric, Δ^{ARE} , in terms of the point-wise mutual information (pmi), which appears naturally in the expressions for various metrics considered in this work. Usually, the pmi is defined as $\log \Lambda$, where $\Lambda : \mathbb{F}_q \otimes \mathcal{L} \rightarrow \mathbb{R}$, is defined as $\Lambda(s, \ell) = \mathbb{P}(\mathbf{L} = \ell \mid S = s) / \mathbb{P}(\mathbf{L} = \ell)$ (Definition 17). Then $\Delta^{\text{ARE}}(S; \mathbf{L}) = \mathbb{E}_{\mathbf{L}} [\max_s |\Lambda - 1|]$, and therefore we are interested in bounding $\max_s |\Lambda - 1|$. However, instead of bounding this expression directly, we bound $\max_s |\Lambda' - 1|$, where $\Lambda'(s, \ell) = \mathbb{P}(\mathbf{L} = \ell \mid S = s) / \prod_{i=1}^n \mathbb{P}(L_i = \ell_i)$. Using the triangle inequality, Corollary 8 then gives us a recipe for bounding $\max_s |\Lambda - 1|$ using bounds on $\max_s |\Lambda' - 1|$. Similar to the approach employed by Benhamouda *et al.* in [7], we then use the Poisson summation formula to express bounds on $|\Lambda' - 1|$ in terms of the threshold of the underlying scheme, and the Fourier biases of the posterior distributions $\mathbb{P}(U_i \mid L_i = \ell_i)$.

Independence of Leakage Channels. Intuitively, the independence of leakage channels should allow us to combine their security guarantees, yielding bounds that improve exponentially with the threshold t . In contrast, consider a leakage function $\mathbf{L} = (U_1 + E, \dots, U_n + E)$, where the leakage channels are correlated; here the overall leakage from the secret is greater than leakage from any single channel, $\text{MI}(S; \mathbf{L}) = \text{MI}(S; (U_1 + E, U_2 - U_1, \dots, U_n - U_1)) \geq \text{MI}(U_1, L_1)$. Therefore, to obtain meaningful security bounds, we must analyze how leakage aggregates across channels. Motivated by this, in Lemma 4, we rewrite $\mathbb{P}(\mathbf{L} = \ell \mid S = s)$ in a form that reduces the problem to analyzing individual channels. Let $\Psi_{\ell_i} : \mathbb{F}_q \rightarrow \mathbb{C}$ denote the marginal probabilities of the leakage channels, i.e. $\Psi_{\ell_i}(\alpha) = \mathbb{P}(U_i = \alpha \mid L_i = \ell_i)$, and let $\Psi_{\ell}(\boldsymbol{\alpha} = (\alpha_1, \dots, \alpha_n)) = \prod_{i=1}^n \Psi_{\ell_i}(\alpha_i)$. Then, after marginalizing over $\mathbf{u}$ and using the Bayes' rule, one can observe that $\Lambda'(s, \ell) = |\mathcal{C}| \cdot \sum_{\mathbf{u} \in \mathbf{v} + \mathcal{C}^\perp} \Psi_{\ell}(\mathbf{u})$, where $\mathcal{C} \subseteq \mathbb{F}_q^n$ denotes the linear code spanned by the coefficients of the secret recovery equations, and $\mathcal{C}^\perp$ its dual.

Poisson Summation. Recall that, for (n, t) -threshold schemes, $\mathcal{C}$ is an $[n, k, t]_q$ code. So far we have neither used the linearity of the deployed scheme, and nor have we used the threshold t . To leverage these code properties, we use the Poisson summation formula, from the theory of Fourier analysis over Abelian groups (see Section 2). This formula allows us to transform sums of the function Ψ_{ℓ} over cosets of the linear code $\mathcal{C}^\perp$, to sums of its Fourier transform, $\widehat{\Psi}_{\ell} = \prod_{i=1}^n \widehat{\Psi}_{\ell_i}(\alpha_i)$, over the code $\mathcal{C}$ itself. This yields the following bound from Lemma 4, $\max_s |\Lambda'(s, \ell) - 1| \leq \sum_{\mathbf{w} \in \mathcal{C}^\times} |\widehat{\Psi}_{\ell}(\mathbf{w})|$, where $\mathcal{C}^\times$ is set of non-zero codewords in $\mathcal{C}$. Then, using the relationship between $\max_{s \in \mathbb{F}_q} |\Lambda - 1|$ and $\max_{s \in \mathbb{F}_q} |\Lambda' - 1|$ from Corollary 8, and taking an expectation over $\mathbf{L}$, we obtain the following inequality $\Delta^{\text{ARE}}(S; \mathbf{L}) \leq 2 \cdot \sum_{\mathbf{w} \in \mathcal{C}^\times} \prod_{i=1}^n \mathbb{E}_{L_i} [|\widehat{\Psi}_{\ell_i}(\mathbf{w})|]$.

Fourier Bias. Note that $|\widehat{\Psi}_{\ell_i}(\alpha_i)| = 1$ if and only if $\alpha_i = 0$. For a function $f : \mathbb{F}_q \rightarrow \mathbb{C}$, the maximum absolute value taken by its Fourier transform,

$\max_{\alpha \in \mathbb{F}_q^\times} |\widehat{f}(\alpha)|$, is called its Fourier linear bias, and is denoted by $\text{bias}(f)$. We extend this notion to random variables via their probability mass functions (Section 3). Then for $\alpha \neq 0$, $|\widehat{\Psi}_{\ell_i}(\alpha)| \leq \text{bias}(U_i | L_i = \ell_i)$. Since the secret shares U_i are distributed uniformly over $\mathbb{F}_q$, using Lemma 3, we have $\mathbb{E}_{L_i}[|\widehat{\Psi}_{\ell_i}(\mathbf{w})|] \leq \mathbb{E}_{L_i}[\text{bias}(U_i | L_i = \ell_i)] = \text{bias}(U_i | L_i) \leq 2\Delta^{\text{TV}}(U_i; L_i) := 2 \cdot \text{TV}(U_i; U_i | L_i)$, where TV is the total variation distance metric.

Main Result. To finally upper bound the required metric, $\Delta^{\text{ARE}}(S; \mathbf{L})$, we will use the fact that the minimum weight of the code $\mathcal{C}$ is t . For some $\mathbf{w} \in \mathcal{C}$, let $\mathcal{J} \subseteq \text{supp}(\mathbf{w})$ be a subset of size t in the support of $\mathbf{w}$. Then $\mathcal{J} \in \binom{[n]}{t}$, and $\prod_{i=1}^n \mathbb{E}_{L_i}[|\widehat{\Psi}_{\ell_i}(\mathbf{w})|] \leq \prod_{j \in \mathcal{J}} \text{bias}(U_j | L_j) \leq 2^t \cdot \max_{\mathcal{J} \in \binom{[n]}{t}} \prod_{j \in \mathcal{J}} \Delta^{\text{TV}}(U_j, L_j)$. Plugging this back, into the inequality with Δ^{ARE} , we get the proven upper bound. The main result of this work, Theorem 4, has been proven in Section 6.

1.4 Outline

The remainder of this paper is structured as follows. In Section 2, we first recall the procedure of constructing threshold secret sharing schemes from error-correcting codes and then discuss some results from discrete Fourier analysis used in this work. In Section 3, we formally define the leakage metrics being considered in this work, including mutual information, total variation distance, and a metric based on Fourier linear bias. In Section 4, we recall the δ -noisy leakage model and some other leakage models which are widely studied in noisy secret sharing. In Section 5, we present the main result of this paper, Theorem 4, and discuss its relevance in designing secret sharing schemes with practical security guarantees. Then in Section 6, we provide a formal mathematical proof of the central result of this paper. Finally, the paper is concluded in Section 7.

2 Preliminaries

We begin by stating the mathematical notations in Section 2.1. We then review the theory of error-correcting codes, and linear secret sharing schemes in Section 2.2. Then in Section 2.3, we recall some definitions and results from the theory of discrete Fourier analysis which will be used in this work, including the Poisson summation formula. The connection between leakage-resilient secret sharing and Fourier analytical methods used in additive combinatorics was first made by Benhamouda *et al.* in Crypto'18 [6]. We refer the reader to the journal version of this work [7] for a detailed discussion.

2.1 Notation

Let $\mathbb{F}_q$ be a finite field of order $q = p^d$, where $p \geq 2$ is a prime and $d \in \mathbb{Z}_+$. Let $\mathbb{F}_q^n$ denote the ambient vector space of vectors of length $n \in \mathbb{Z}_+$ over the field $\mathbb{F}_q$. The Hamming weight of a vector $\mathbf{x} \in \mathbb{F}_q^n$, denoted by $\text{wt}(\mathbf{x})$, is defined

as the number of non-zero coordinates in the standard representation of $\mathbf{x}$, i.e., $wt(\mathbf{x} = (x_1, \dots, x_n)) = \#\{x_j : x_j \neq 0, 1 \leq j \leq n\}$. We will use the standard inner product notation $\langle \cdot, \cdot \rangle$ to represent the inner product of two vectors over $\mathbb{F}_q$. For $\mathbf{x} = (x_1, \dots, x_n) \in \mathbb{F}_q^n$ and $\mathbf{y} = (y_1, \dots, y_n) \in \mathbb{F}_q^n$, their inner product is defined as $\langle \mathbf{x}, \mathbf{y} \rangle = x_1 y_1 + \dots + x_n y_n$. The sets of values taken by random variables in this work will be discrete, and usually belong to $\mathbb{F}_q$. Random variables are denoted by uppercase letters, e.g. X , and their realization will be denoted by the corresponding lowercase letter $x \leftarrow X$. For some discrete random variable X , its probability mass function (pmf) will be written as $p_X(x) = \mathbb{P}(X = x)$. Random vectors are denoted by boldface uppercase letters, e.g. $\mathbf{X} = (X_1, \dots, X_n)$, and their realization will be denoted by the corresponding boldface lowercase letter $\mathbf{x} = (x_1, \dots, x_n) \leftarrow \mathbf{X}$. For some random vector $\mathbf{X}$, its probability mass function (pmf) will be written as $p_{\mathbf{X}}(\mathbf{x}) = \mathbb{P}(\mathbf{X} = \mathbf{x})$. For a vector $\mathbf{x} = (x_1, \dots, x_n) \in \mathbb{F}_q^n$ and a random vector $\mathbf{V} = (V_1, \dots, V_n)$ taking values in $\mathbb{F}_q^n$, we will extend this notation to represent a linear combination of random variables as $\langle \mathbf{x}, \mathbf{V} \rangle = x_1 V_1 + \dots + x_n V_n$, which is a random variable taking values in $\mathbb{F}_q$. For such a linear combination of random variables, the vector $\mathbf{x}$ is called the coefficient vector corresponding to equation $\langle \mathbf{x}, \mathbf{V} \rangle$. And lastly, the transpose of a matrix $A \in \mathbb{F}_q^{m \times n}$ will be denoted by $A^\top \in \mathbb{F}_q^{n \times m}$.

2.2 Codes and Linear Secret Sharing Schemes

It is well-known that several linear secret-sharing schemes can be constructed from linear error-correcting codes [11, 13, 37, 38, 40]. Massey utilized linear codes for secret sharing schemes and established the relationship between the access structure and the minimal codewords of the dual code [37, 38]. The seminal Shamir's scheme [47] is a specific instance of Massey's secret sharing scheme, where the underlying construction relies upon Reed-Solomon codes [40]. We begin by reviewing the definition of linear error-correcting codes and dual codes.

Linear codes. A linear code $\mathcal{C}$ of length n over $\mathbb{F}_q$ is a subspace of the ambient vector space $\mathbb{F}_q^n$. Recall that weight of a vector $\mathbf{u} \in \mathbb{F}_q^n$, $wt(\mathbf{u})$, is the number of non-zero coordinates of $\mathbf{u}$. The minimum distance of a linear code is the smallest non-zero weight of a vector in $\mathcal{C}$. The notation $[n, k, d]_q$ represents a linear code of length n , dimension k , and minimum distance d , over the finite field $\mathbb{F}_q$. Vectors belonging to some code $\mathcal{C}$ are also called *codewords*.

Definition 1. An $[n, k, d]_q$ code is a k -dimensional subspace $\mathcal{C} \subseteq \mathbb{F}_q^n$, with minimum distance d , where $d = \min\{wt(\mathbf{c}) : \mathbf{c} \in \mathcal{C}, \mathbf{c} \neq \mathbf{0}\}$.

A generator matrix of an $[n, k, d]_q$ code $\mathcal{C}$ is a matrix in $\mathbb{F}_q^{k \times n}$ whose rows span the subspace $\mathcal{C} \subseteq \mathbb{F}_q^n$. We abuse notation and use the same calligraphic letter $\mathcal{C}$ to denote the generator matrix of the corresponding code. Any codeword $\mathbf{c}$ of this codebook can be written as $\mathbf{c} = \mathbf{u}\mathcal{C}$ for some row vector $\mathbf{u} \in \mathbb{F}_q^k$.

Definition 2. The dual of an $[n, k, d]_q$ code $\mathcal{C}$ is an $[n, n - k, d^\perp]_q$ code $\mathcal{C}^\perp$, defined as the orthogonal complement of $\mathcal{C}$ in $\mathbb{F}_q^n$. Formally,

$$\mathcal{C}^\perp = \{\mathbf{v} \in \mathbb{F}_q^n : \langle \mathbf{v}, \mathbf{c} \rangle = 0 \text{ for all } \mathbf{c} \in \mathcal{C}\},$$

where $\langle \cdot, \cdot \rangle$ denotes the standard inner product over $\mathbb{F}_q$. The dual code $\mathcal{C}^\perp$ is a linear code of dimension $n - k$ and minimum distance $d^\perp$.

Massey's Secret Sharing. Given a secret $s \in \mathbb{F}_q$ and a threshold $t \in \mathbb{Z}_+$, the Massey's secret-sharing scheme distributes secret shares $u_1, u_2, \dots, u_n \in \mathbb{F}_q$ among n users such that:

- the secret s is a linear combination of at least t shares;
- fewer than t shares reveal no information about the secret.

Such a scheme is constructed using an $[n+1, k+1]_q$ linear code $\mathcal{M} \subseteq \mathbb{F}_q^{n+1}$, where the dual code $\mathcal{M}^\perp$ is an $[n+1, n-k, t+1]_q$ code. For a secret $s \leftarrow S$, the scheme samples a uniformly random codeword $\mathbf{c} = (u_0, \dots, u_n) \in \mathcal{M}$ such that $u_0 = s$. The i -th share is defined as $u_i \in \mathbb{F}_q$ for $i \in \{1, 2, \dots, n\}$. Massey established the connection between the access structure of the secret-sharing scheme and the minimal codewords of the dual code $\mathcal{M}^\perp$ [37, 38]. Specifically, they showed that there exists a vector $\mathbf{v} = (v_0, \dots, v_n) \in \mathcal{M}^\perp$, such that $v_0 \neq 0$ and $\text{wt}(\mathbf{v}) = t+1$. Since, such a vector $\mathbf{v}$ is orthogonal to $\mathbf{c} \in \mathcal{M}$, i.e., $\langle \mathbf{c}, \mathbf{v} \rangle = 0$; the threshold of the scheme is t . A Massey secret-sharing scheme with n users, threshold t , and constructed using code $\mathcal{M} \subseteq \mathbb{F}_q^{n+1}$, is denoted by $\text{MasseySS}(n, t, \mathcal{M})$.

Let the secret shares' vector be denoted by $\mathbf{u} = (u_1, \dots, u_n) \leftarrow \mathbf{U}$. This vector is formed by deleting the first coordinate in each codeword of $\mathcal{M}$. The resulting code $\mathcal{U} \subseteq \mathbb{F}_q^n$ is a linear code of length n . We first begin by showing that for a given secret s , all corresponding secret shares' vectors are equiprobable. This follows from the fact that the Massey's scheme samples a uniformly random codeword $(u_0, \dots, u_n) \in \mathcal{M}$ such that $u_0 = s$.

Proposition 1. *If two secret shares' vectors $\mathbf{u}$ and $\mathbf{u}'$ are such that both of them correspond to the same secret $s \in \mathbb{F}_q$, i.e. $\mathbf{u}\mathcal{C}_{\mathcal{M}}^\top = \mathbf{u}'\mathcal{C}_{\mathcal{M}}^\top$, then $p_{\mathbf{U}}(\mathbf{u}) = p_{\mathbf{U}}(\mathbf{u}')$.*

Proof. For some secret shares' vector $\mathbf{u}$ corresponding to some secret s , the coefficients $\mathbf{p} = (s, p_1, \dots, p_{t-1}) \leftarrow \mathbf{P}$ are uniquely determined. Since the random variables S , and P_i 's are independent, $p_{\mathbf{U}}(\mathbf{u}) = p_S(s) \prod_{i=1}^{t-1} p_{P_i}(p_i) = p_S(s)q^{1-t}$.

Definition 3. *Given $\mathbf{v} = (v_1, \dots, v_n) \in \mathbb{F}_q^n$, a linear combination of the secret shares $\langle \mathbf{v}, \mathbf{U} \rangle = v_1U_1 + \dots + v_nU_n$ is called a secret recovery equation if*

$$S = \langle \mathbf{v}, \mathbf{U} \rangle. \quad (1)$$

Given the $\text{MasseySS}(n, t, \mathcal{M})$ scheme with threshold t , each secret recovery equation has at least t non-zero summands. Therefore, for every coefficient vector $\mathbf{v} \in \mathbb{F}_q^n$ corresponding to a secret recovery equation, we have $\text{wt}(\mathbf{v}) \geq t$. An adversary with knowledge of the underlying codebook $\mathcal{M}$ has access to the coefficients of all secret recovery equations, and consequently, the linear combinations of these equations. The linear subspace spanned by these coefficient vectors is denoted as $\mathcal{C}_{\mathcal{M}}$. This subspace $\mathcal{C}_{\mathcal{M}}$ is a linear code of length n over $\mathbb{F}_q$.

Definition 4. Let $\mathcal{V}$ be the set of coefficient vectors corresponding to all possible secret recovery equations, i.e., $\mathcal{V} = \{\mathbf{v} \in \mathbb{F}_q^n : \langle \mathbf{v}, \mathbf{U} \rangle = S\}$. Let $\mathcal{C}_{\mathcal{M}}$ be the subspace of $\mathbb{F}_q^n$ spanned by the vectors in $\mathcal{V}$,

$$\mathcal{C}_{\mathcal{M}} = \text{span}(\mathcal{V}) = \left\{ \sum_{i=1}^k \alpha_i \mathbf{v}_i : \mathbf{v}_i \in \mathcal{V}, \alpha_i \in \mathbb{F}_q, k \in \mathbb{Z}_+ \right\}. \quad (2)$$

Lemma 1 states that the minimum distance of the code $\mathcal{C}_{\mathcal{M}}$ is equal to the threshold t . Using the singleton bound, we find that the dimension of $\mathcal{C}_{\mathcal{M}}$ satisfies $k \leq n - t + 1$. In Section 6, we formally show how the bounds on the security metrics, for the information leaked about the secret, depend on the threshold t .

Lemma 1. The code $\mathcal{C}_{\mathcal{M}}$ is an $[n, k, d]_q$ code with $k \leq n - t + 1$, and $d = t$.

2.3 Fourier Analysis

In this section, we present the notion of Fourier transform of a function and some of its properties. Let $\mathbb{C}$ be the field of complex numbers and let $\mathbb{F}_q$ be a finite field extension of $\mathbb{F}_p$, where $q = p^d$. The Trace map is the function $\text{Tr} : \mathbb{F}_q \rightarrow \mathbb{F}_p$ defined by $\text{Tr}(\alpha) = \alpha + \alpha^p + \alpha^{p^2} + \cdots + \alpha^{p^{d-1}}$. Let α_i denote the elements of $\mathbb{F}_q$, i.e., $\mathbb{F}_q = \{\alpha_0, \dots, \alpha_{q-1}\}$, such that $\alpha_0 = 0$.

Definition 5 (Character Function). Let $\omega = e^{2\pi i/p}$ be a primitive p^{th} root of unity. The canonical additive character $\chi : \mathbb{F}_q \rightarrow \mathbb{C}$ is defined as $\chi(\alpha) = \omega^{\text{Tr}(\alpha)}$.

The theory of additive character functions is more general, applying to any finite Abelian group $(\mathbb{G}, +)$. A character is a homomorphism $\chi : \mathbb{G} \rightarrow \mathbb{C}^\times$ from the group $\mathbb{G}$ to non-zero complex numbers $\mathbb{C}^\times$, i.e., $\chi(\alpha + \beta) = \chi(\alpha) \cdot \chi(\beta)$, such that $|\chi(\alpha)| = 1$, for all $\alpha \in \mathbb{G}$. For any finite Abelian group $\mathbb{G}$, the set of characters $\hat{\mathbb{G}}$ is a group (under the operation point-wise product) isomorphic to $\mathbb{G}$. The scope of this work, however, will be confined to the specific and important case where $\mathbb{G}$ is the additive group of a finite field, $(\mathbb{F}_q, +)$.

Definition 6 (Fourier Transform). For functions $f : \mathbb{F}_q^n \rightarrow \mathbb{C}$, we define the Fourier transform $\hat{f} : \mathbb{F}_q^n \rightarrow \mathbb{C}$ as

$$\hat{f}(\boldsymbol{\alpha}) = \sum_{\mathbf{x} \in \mathbb{F}_q^n} f(\mathbf{x}) \cdot \chi(\langle \boldsymbol{\alpha}, \mathbf{x} \rangle), \quad (3)$$

where $\langle \boldsymbol{\alpha}, \mathbf{x} \rangle$ denotes the standard inner product in $\mathbb{F}_q^n$ and $\chi(z) = \omega^{\text{Tr}(z)}$ is the canonical additive character.

A key result from Fourier analysis is the Poisson summation formula, which relates the sum of a function over a subspace to the sum of its Fourier transform over the dual subspace. This principle has been widely applied in fields like number theory and additive combinatorics. Of particular relevance to our work, Benhamouda *et al.* employed the version for linear codes [6, 7] to show the security of (n, t) -threshold schemes with bounded leakage.

Theorem 2 (Poisson Summation, [7], Lemma 4.16). *Let $\mathcal{C}$ be an $[n, k, d]_q$ linear code with dual $\mathcal{C}^\perp$. For any function $f : \mathbb{F}_q^n \rightarrow \mathbb{C}$ and any $\mathbf{v} \in \mathbb{F}_q^n$,*

$$\sum_{\mathbf{u} \in \mathbf{v} + \mathcal{C}} f(\mathbf{u}) = \frac{1}{|\mathcal{C}^\perp|} \sum_{\mathbf{w} \in \mathcal{C}^\perp} \widehat{f}(\mathbf{w}) \chi(-\langle \mathbf{w}, \mathbf{v} \rangle). \quad (4)$$

We now introduce the notion of Fourier linear bias, which characterizes the deviation of a function from the uniform distribution. Intuitively, a function is more biased if it is highly correlated with some Fourier character. For binary extension fields, the bias has been widely used in Boolean analysis (see [44]).

Definition 7 (Bias of a Function). *For a function $f : \mathbb{F}_q \rightarrow \mathbb{C}$, the bias of f is defined as*

$$\text{bias}(f) = \max_{\alpha \in \mathbb{F}_q^\times} |\widehat{f}(\alpha)|, \quad (5)$$

where $\mathbb{F}_q^\times = \mathbb{F}_q \setminus \{0\}$ denotes the multiplicative group of the field $\mathbb{F}_q$.

In Section 3, we will extend the definition of Fourier bias to random variables. In the proof of our main result, we will first use the Poisson summation formula to translate the problem of bounding distance metrics, to bounding sums of Fourier transforms. Then, we will be able to use the bias of the random variable to bound this summation. Lemma 3 allows us to bound the Fourier bias, in terms of other relevant security metrics being considered in this work (Section 3).

3 Information Leakage Measures

We start with some standard definitions and results about mutual information (MI) and the total variation distance (TV) metrics. A key advantage of such statistical metrics is their universality, as they assume an adversary with unbounded computational power, providing an upper bound on the information leakage irrespective of implementation or attack complexity. Subsequently, we introduce new metrics, based on the Fourier linear bias, which we will later use in Section 6, to prove the main result of this work, Theorem 4 (Section 5).

3.1 Entropy and Mutual Information

Let X be a discrete random variable with probability mass function $p_X(x)$. The base- q entropy of X , denoted by $H_q(X)$, quantifies the uncertainty of X and is defined as $H_q(X) = -\sum_x p_X(x) \log_q p_X(x)$, where the logarithm is taken to base q . For $q = 2$, the base is typically omitted, and $H_q(X)$ is simply written as $H(X)$, commonly measured in bits. The relationship between the two is given by $H(X) = \log_2(q) \cdot H_q(X)$. Mutual information between two random variables measures the average reduction in uncertainty about X given knowledge of another random variable L . For two random variables X and Y , the base- q mutual information, $\text{MI}_q(X; L)$, is defined as:

$$\text{MI}_q(X; L) = H_q(X) - H_q(X/L). \quad (6)$$

For $q = 2$, we again omit the base, and $\text{MI}(X; L) = \log_2(q) \cdot \text{MI}_q(X; L)$.

3.2 Total Variation Distance

The total variation distance, also known as statistical distance, measures the distinguishability between two probability distributions. If A and B are random variables, over the same alphabet $\mathcal{X}$, the total variation distance metric is defined as $\text{TV}(A; B) = \frac{1}{2} \sum_{x \in \mathcal{X}} |\mathbb{P}(A = x) - \mathbb{P}(B = x)|$. Since the TV metric only depends upon the probability distribution of the random variables, when clear from context, we will use the notation $\text{TV}(p_A; p_B) = \text{TV}(A; B)$. The TV metric is further used to define several other metrics. The *pointwise bias* is a metric used to determine the advantage an adversary gains about the secret X after observing a single, specific leakage outcome $\ell \leftarrow L$.

Definition 8 (Pointwise TV Bias). *Given a discrete random variable X over $\mathcal{X}$ and some given leakage outcome $\ell \in \mathcal{L}$, the pointwise TV bias is defined as*

$$\Delta_\ell^{\text{TV}}(X; L) \triangleq \text{TV}(p_{X|L=\ell}; p_X) = \frac{1}{2} \sum_{x \in \mathcal{X}} |\mathbb{P}(X = x | L = \ell) - \mathbb{P}(X = x)|.$$

The overall leakage of the system is the average of the leakages from each specific observation. The *statistical TV bias* quantifies the average TV-metric based information that some side-channel leakage L reveals about a secret X .

Definition 9 (Statistical TV Bias). *The statistical bias between two random variables X and L , over alphabets $\mathcal{X}$ and $\mathcal{L}$ respectively, is defined as*

$$\Delta^{\text{TV}}(X; L) \triangleq \mathbb{E}_L[\Delta_\ell^{\text{TV}}(X; L)] = \text{TV}(p_{XL}; p_X \otimes p_L).$$

Expanding, this gives us $\Delta(X; L) = \frac{1}{2} \sum_{x \in \mathcal{X}} \sum_{\ell \in \mathcal{L}} |\mathbb{P}(X = x, L = \ell) - \mathbb{P}(X = x)\mathbb{P}(L = \ell)|$. Furthermore, observe that $\Delta(X; L) = \sum_{\ell \in \mathcal{L}} \mathbb{P}(L = \ell) \cdot \Delta_\ell(X; L) \leq \max_{\ell \in \mathcal{L}} \Delta_\ell(X; L)$. The Pinsker's inequality relates the TV and MI metrics.

Theorem 3 (Pinsker's Inequality). *For discrete random variables X and L ,*

$$\Delta^{\text{TV}}(X; L) \leq \sqrt{\frac{1}{2} \ln 2 \cdot \text{MI}(X; L)}.$$

3.3 Other Leakage Metrics

Motivated by their use in bounding the MI and Δ^{TV} metrics, several other ad hoc metrics to the statistical bias have been studied in the cryptanalysis literature. In this work, we will additionally consider the Average Relative Error metric [45, Definition 3] and the Complementary Doeblin Coefficients [3, Definition 9].

Definition 10 (Complementary Doeblin Coefficient). *The complementary Doeblin coefficient between two random variables X and L , over alphabets $\mathcal{X}$ and $\mathcal{L}$ is defined as*

$$\Delta^{\text{CDC}}(X; L) \triangleq \mathbb{E}_L \left[\sup_{x \in \mathcal{X}} \left(1 - \frac{\mathbb{P}(X = x | L = \ell)}{\mathbb{P}(X = x)} \right) \right].$$

Definition 11 (Average Relative Error). *The average relative error between two random variables X and L , over alphabets $\mathcal{X}$ and $\mathcal{L}$ is defined as*

$$\Delta^{\text{ARE}}(X; L) \triangleq \mathbb{E}_L \left[\sup_{x \in \mathcal{X}} \left| \frac{\mathbb{P}(X = x \mid L = \ell)}{\mathbb{P}(X = x)} - 1 \right| \right].$$

It is easy to see that $\Delta^{\text{CDC}}(X; L) \leq \Delta^{\text{ARE}}(X; L)$. The Δ^{ARE} metric can be used to upper bound both the Δ^{TV} [45, Proposition 1] and MI [3, Lemma 9] metrics. In Section 6, we will show a bound on the overall leakage for the Δ^{ARE} metric, which will therefore imply the security for other metrics discussed in this section.

Lemma 2 ([3], Lemma 9). *For discrete random variables X and L , $\Delta^{\text{TV}}(X; L) \leq \frac{1}{2} \Delta^{\text{ARE}}(X; L)$, and $\text{MI}(X; L) \leq H(X) \cdot \Delta^{\text{ARE}}(X; L)$.*

3.4 Fourier Linear Bias

Here we introduce new metrics based on the notion of Fourier bias (Definition 7). For a random variables X taking values in $\mathbb{F}_q$, its Fourier bias is the bias of the corresponding probability mass function. Then similar to pointwise and statistical TV biases (Definitions 8 and 9), we use the Fourier bias to define the pointwise and statistical FL (Fourier Linear) biases.

Definition 12 (Fourier Linear Bias of a Random Variable). *Let X be a random variable taking values in $\mathbb{F}_q$, and let $p_X : \mathbb{F}_q \rightarrow \mathbb{R}$ be its probability mass function with $p_X(x) := \Pr[X = x]$. The bias of X is defined as*

$$\text{bias}(X) := \text{bias}(p_X) = \max_{\alpha \in \mathbb{F}_q^\times} |\mathbb{E}_X[\chi(\alpha X)]|.$$

Definition 13 (Pointwise FL bias). *Given a random variable X over $\mathbb{F}_q$ and some leakage outcome $\ell \in \mathcal{L}$, the pointwise Fourier linear bias is defined as*

$$\Delta_\ell^{\text{FL}}(X; L) \triangleq \frac{1}{2} \max_{\alpha \in \mathbb{F}_q^\times} |\mathbb{E}_X[\chi(\alpha X) \mid L = \ell] - \mathbb{E}_X[\chi(\alpha X)]|.$$

Using the triangle inequality, $\Delta_\ell^{\text{FL}}(X; L) \leq \frac{1}{2}(\text{bias}(X) + \text{bias}(X \mid L = \ell))$. When X is uniformly distributed, $\mathbb{E}_X[\chi(\alpha X)] = 0$; therefore $\text{bias}(X) = 0$ and $\Delta_\ell^{\text{FL}}(X; L) = \frac{1}{2} \text{bias}(X \mid L = \ell)$. If X is determined uniquely by ℓ , then $\text{bias}(X \mid L = \ell) = 1$.

Definition 14 (Statistical FL bias). *Given a random variable X over $\mathbb{F}_q$, and leakage variable L , the statistical Fourier linear bias is defined as*

$$\Delta^{\text{FL}}(X; L) \triangleq \mathbb{E}_L[\Delta_\ell^{\text{FL}}(X; L)] = \frac{1}{2} \mathbb{E}_L \left[\max_{\alpha \in \mathbb{F}_q^\times} |\mathbb{E}_X[\chi(\alpha X) \mid L = \ell] - \mathbb{E}_X[\chi(\alpha X)]| \right].$$

In the main results of this work, Theorem 4, we show upper bounds for the leakage in terms of the statistical FL bias metric, Δ^{FL} . Lemma 3 shows how the statistical FL bias is bounded by other security metrics discussed in this section.

Lemma 3. *For a random variable X over $\mathbb{F}_q$ and leakage outcome $L = \ell$, we have $\Delta_\ell^{\text{FL}}(X; L) \leq \Delta_\ell^{\text{TV}}(X; L)$, and consequently $\Delta^{\text{FL}}(X; L) \leq \Delta^{\text{TV}}(X; L) \leq \min \left\{ \left(\frac{\log_e 2 \cdot \text{MI}(X; L)}{2} \right)^{1/2}, \frac{1}{2} \Delta^{\text{ARE}}(X; L) \right\}$. If X is uniformly distributed, then*

$$\Delta^{\text{FL}}(X; L) \leq \min \left\{ \Delta^{\text{TV}}(X; L), \frac{1}{2} \Delta^{\text{CDC}}(X; L) \right\} \leq \frac{1}{2} \Delta^{\text{ARE}}(X; L).$$

Proof. For some leakage $L = \ell$, note that $|\mathbb{E}_X[\chi(\alpha X) | L = \ell] - \mathbb{E}_X[\chi(\alpha X)]| = |\sum_x \chi(\alpha x)(\Pr(X = x | L = \ell) - \Pr(X = x))| \leq \sum_x |\Pr(X = x | L = \ell) - \Pr(X = x)| = 2\Delta_\ell(X; L)$. Hence $\Delta_\ell^{\text{FL}}(X; L) \leq \Delta_\ell^{\text{TV}}(X; L)$, and taking an expectation over L gives $\Delta^{\text{FL}}(X; L) \leq \Delta^{\text{TV}}(X; L)$. Now using Theorem 3 and Lemma 2, we get the first inequality. When X is uniform, $\mathbb{E}_X[\chi(\alpha X)] = 0$. For some given $L = \ell$, let $p_{\min} = \min_x \Pr(X = x | L = \ell)$. Then $|\mathbb{E}_X[\chi(\alpha X) | L = \ell] - \mathbb{E}_X[\chi(\alpha X)]| = |\mathbb{E}_X[\chi(\alpha X) | L = \ell] - p_{\min} \cdot \sum_x \chi(\alpha x)| \leq |\sum_x \Pr(X = x | L = \ell) - p_{\min}| \leq 1 - q \cdot p_{\min}$. Taking an expectation over L , we are done.

From Lemma 3, we see that the Δ^{FL} metric is smaller (with constant factor reductions) than the metrics $\{\Delta^{\text{TV}}, \Delta^{\text{CDC}}, \text{MI}^{1/2}, \Delta^{\text{ARE}}\}$. Therefore, in Section 6, we bound the information leaked about the secret in terms of the leakage from each secret share for the Δ^{FL} metric, which will imply security for the others. The security parameter $\eta_{\mathcal{M}}$ (Definition 16) used to characterize the bounds main results of this work is also therefore defined using this metric (see Section 5).

4 Leakage Model

In this section, we will first recall the δ -noisy leakage model proposed by Prouff and Rivain in Eurocrypt'13 [46]. The model assumes that the adversary receives a noisy version of every secret share, with some minimum distortion. We then briefly discuss the random probing (RP) model [14] which has been widely studied in the security analysis of (n, n) -threshold schemes as an intermediary model to bridge the limitations of theoretical techniques, and real-world leakage models.

4.1 δ -Noisy Leakage Model

In the δ -noisy leakage model, each secret share leaks information to the adversary with some leakage parameter δ . Different works have considered different metrics to quantify this parameter, with the most popular being the Δ^{TV} and MI metrics. Formally, for some leakage metric Δ , we define the following.

Definition 15 (δ -Noisy Channel). *For some random variable X over $\mathcal{X}$, a (randomized) leakage function $L : \mathcal{X} \rightarrow \mathcal{L}$ is said to be δ -noisy leakage channel if $\Delta(X; L) \leq \delta$. The leakage output will be called a δ -noisy observation of X .*

Consider an adversary who for some metric Δ^{sha} , observes a δ -noisy version of every secret share, and wants to compromise the security of the secret. Formally,

consider the Massey's secret sharing scheme, $\text{MasseySS}(n, t, \mathcal{M})$, with n users, and threshold t , over a finite field $\mathbb{F}_q$. Let $s \leftarrow S$ denote the secret that is shared, and $\mathbf{u} = (u_1, \dots, u_n) \leftarrow \mathbf{U}$ denote the secret shares' vector. The leaked version of the secret shares' vector will be denoted by $\boldsymbol{\ell} = (\ell_1, \dots, \ell_n) \leftarrow \mathbf{L}$. The leakage model we consider has the following assumptions,

- the adversary obtains a δ -noisy observation of every secret share, i.e. for the i^{th} secret share, $\Delta^{\text{sha}}(U_i; L_i) \leq \delta_i$,
- information leaked to the adversary from each user is independent of the leakage from other users, i.e. the channels $L_i \mid U_i$ are mutually independent.

In Section 6, we prove our results for the $\Delta^{\text{sha}} = \Delta^{\text{FL}}$ metric. Recall from Lemma 3 that the statistical FL bias metric (Definition 14) is such that $\Delta^{\text{FL}} \leq \Delta^{\text{sha}}$, for $\Delta^{\text{sha}} \in \{\Delta^{\text{FL}}, \Delta^{\text{TV}}, \frac{\Delta^{\text{CDC}}}{2}, \left(\frac{\log_e 2 \cdot \text{MI}}{2}\right)^{1/2}, \frac{\Delta^{\text{ARE}}}{2}\}$.

4.2 Noisy Leakage Models in Additive Secret Sharing Schemes

A large body of work has studied the security of additive secret sharing schemes, i.e. (n, n) -threshold schemes and circuit compilers for various noisy leakage models [2, 3, 4, 9, 12, 15, 17, 18, 19, 20, 24, 25, 29, 39, 43, 45]. Several prior works have considered proving bounds on the security guarantees for the δ -noisy leakage model. For general settings, the δ -noisy leakage model is often reduced to other more analytically tractable leakage models, on which security guarantees can be proved more easily. The first such result was shown by Duc *et al.* in Eurocrypt'14, for the random probing (RP) model [14].

ϵ_{RP} -Random Probing Model. In this model, each intermediate value $X \in \mathbb{F}_q$ is either fully revealed to the adversary with probability $\epsilon_{\text{RP}} \in [0, 1)$ or remains hidden. Formally, for a $\text{MasseySS}(n, t, \mathcal{M})$ scheme over $\mathbb{F}_q$, the leakage L_i for each secret share U_i is defined as

$$\ell_i = \begin{cases} u_i & \text{with probability } \epsilon_{\text{RP}}, \\ \perp & \text{with probability } 1 - \epsilon_{\text{RP}}. \end{cases} \quad (7)$$

Duc *et al.* proved that the δ -noisy leakage model can be perfectly simulated by the ϵ_{RP} -random probing model, provided that $\epsilon_{\text{RP}} = \delta \cdot |\mathcal{X}|$, where $\mathcal{X}$ is the set of all values that any wire may take in the circuit, *e.g.*, the field size in an arithmetical circuit. In Section 5, we show bounds on the security of (n, t) -threshold schemes with the ϵ_{RP} -random probing leakage using the following proposition.

Proposition 2. *Let X be uniformly distributed over $\mathbb{F}_q$, and let L denote the leakage function under the ϵ_{RP} -random probing model. Then $2\Delta^{\text{FL}}(X; L) = \epsilon_{\text{RP}}$.*

Proof. Since X is uniformly distributed, $\mathbb{E}_X[\chi(\alpha X)] = 0$ and therefore $2\Delta^{\text{FL}}(X; L) = \mathbb{E}_X[\chi(\alpha X) | L = \ell]$. For any $\alpha \neq 0$, if $L = X$, the inner expectation satisfies $|\mathbb{E}_X[\chi(\alpha X) \mid X]| = 1$. If $L = \perp$, the orthogonality of characters implies $\mathbb{E}_X[\chi(\alpha X)] = 0$. Summing over the leakage outcomes:

$$\mathbb{E}_L[|\mathbb{E}_X[\chi(\alpha X) \mid L]|] = \epsilon_{\text{RP}} \cdot 1 + (1 - \epsilon_{\text{RP}}) \cdot 0 = \epsilon_{\text{RP}}.$$

Therefore, we have that the metric $2\Delta^{\text{FL}}(X; L) = \epsilon_{\text{RP}}$.

4.3 Leakage Models in (n, t) -threshold Schemes

Substantial research has been conducted in recent years to analyze the security of (n, t) -threshold schemes against various models of leakage attacks [1, 6, 7, 22, 28, 30, 31, 32, 33, 34, 35, 42]. A majority of these works evaluate security guarantees for the bounded leakage model, first studied by Benhamouda *et al.* in Crypto’18. In this model, the adversary is allowed to learn arbitrary bounded-length information about secret shares, i.e. some function $\tau_i : \mathbb{F}_q \rightarrow \{0, 1\}^l$ from the secret shares of each user, i.e., for each user i , $\tau_i(u_i) \in \{0, 1\}^l$. Such models characterize worst-case leakage and do not directly capture real-world side-channel attacks [48, 49]. This results in conservative security guarantees and that too with severe constraints on the underlying field. In another line of work, Brian *et al.* [10] show how to simulate noisy leakage models from bounded leakage. But the obtained parameters are not good enough to show security for secret sharing schemes with δ -noisy leakage for the MI and Δ^{TV} metrics.

The bounded leakage model has been motivated by capturing wire probing side-channel attacks, whose framework was first introduced by Ishai *et al.* [23]. Such attacks are discrete in nature, and allow adversaries to observe intermediate values during computation by targeting specific hardware wires. The works of Benhamouda *et al.* [6], and Nielsen and Simkin [42], assume each share leaks information independently through arbitrary leakage functions with bounded output-length. Maji *et al.* [30], and Adams *et al.* [1], consider probing attacks which leak physical-bits from the memory hardware storing these shares. In a separate work [31], Maji *et al.* consider joint leakage, where the adversary can leak any bounded output-length joint function of the shares. Some recent studies have focused on understanding the security of (n, t) -schemes under noisy leakage [1, 20, 22]. However, the leakage models considered in these works are ad-hoc noise models which make security guarantees easier to analyze mathematically, but their alignment with real-world side-channel attacks, or their more general theoretical relevance remains unclear.

5 Main Results

In this section, we will state the main result of this work, Theorem 4, where we show bounds on security of (n, t) -threshold schemes with δ -noisy leakage. We assume that each share leaks information to the adversary through independent δ_i -noisy side-channels, i.e. with leakage parameters δ_i . To characterize these guarantees, we define a security parameter $\eta_{\mathcal{M}}$, which only relies upon the threshold t , and the leakage δ_i ’s. Subsequently, we will discuss the practical implications of our result on the design choices for implementing provably secure secret sharing schemes. We will begin by defining the security parameter $\eta_{\mathcal{M}}$.

5.1 Security Parameter $\eta_{\mathcal{M}}$

Formally, for some leakage metric Δ^{sha} , consider the δ -noisy leakage model for Massey’s secret sharing scheme, $\text{MasseySS}(n, t, \mathcal{M})$, where leakage from each

share is bounded as $\Delta^{\text{sha}}(U_i; L_i) \leq \delta_i$. From Section 4.1, recall that for some given channel leakage parameter δ , proving bounds on leakage for the Fourier metric, Δ^{FL} , implies security guarantees (with constant factor reductions) for the metrics $\Delta^{\text{sha}} \in \{\Delta^{\text{FL}}, \Delta^{\text{TV}}, \Delta^{\text{CDC}}, \text{MI}^{1/2}, \Delta^{\text{ARE}}\}$. Therefore, we will use the Δ^{FL} metric, to define the security parameter $\eta_{\mathcal{M}}$.

Definition 16 (Security parameter $\eta_{\mathcal{M}}$). For a MasseySS($n, t, \mathcal{M}$) scheme over $\mathbb{F}_q$ with the δ -noisy leakage model, and leakage variable $\mathbf{L}$, define

$$\eta_{\mathcal{M}} = 2 \cdot \max_{\mathcal{J} \in \binom{[n]}{t}} \left(\prod_{j \in \mathcal{J}} \Delta^{\text{FL}}(U_j; L_j) \right)^{1/t} \leq 2 \cdot \max_{\mathcal{J} \in \binom{[n]}{t}} \left(\prod_{j \in \mathcal{J}} \delta_j \right)^{1/t}, \quad (8)$$

where $\binom{[n]}{t}$ denotes the collection of all t -element subsets of $\{1, \dots, n\}$.

Since the secret shares U_i are uniformly distributed, $2\Delta^{\text{FL}}(U_j; L_j) = \text{bias}(U_j|L_j) \leq 1$. Moreover, from Lemma 3, $2\Delta^{\text{FL}}(U_j; L_j) \leq \Delta^{\text{CDC}}(U_j; L_j) \leq 1$. Therefore, the security parameter $\eta_{\mathcal{M}}$ satisfies the following inequality.

Proposition 3. For some leakage metric Δ^{sha} , consider a MasseySS($n, t, \mathcal{M}$) scheme over $\mathbb{F}_q$ with the δ -noisy leakage model, and security parameter $\eta_{\mathcal{M}}$. Then

$$\eta_{\mathcal{M}} \leq \min \left\{ 1, 2 \cdot \max_{\mathcal{J} \in \binom{[n]}{t}} \left(\prod_{j \in \mathcal{J}} \Delta^{\text{sha}}(U_j; L_j) \right)^{1/t} \right\}, \quad (9)$$

where $\Delta^{\text{sha}} \in \{\Delta^{\text{FL}}, \Delta^{\text{TV}}, \frac{\Delta^{\text{CDC}}}{2}, \left(\frac{\log_e 2 \cdot \text{MI}}{2}\right)^{1/2}, \frac{\Delta^{\text{ARE}}}{2}\}$ and $\binom{[n]}{t}$ denotes the collection of all t -element subsets of $\{1, \dots, n\}$.

5.2 Security of (n, t) -Threshold Schemes

To bound the information leaked from secret S for some metric Δ^{sec} , i.e. bound $\Delta^{\text{sec}}(S; \mathbf{L})$; Lemmas 2 and 3 show that bounding Δ^{ARE} implies security (with a constant reduction factor) for any $\Delta^{\text{sec}} \in \{\Delta^{\text{FL}}, \Delta^{\text{TV}}, \Delta^{\text{CDC}}, \text{MI}, \Delta^{\text{ARE}}\}$. Therefore, in Section 6, we will prove our result for $\Delta^{\text{sec}} = \frac{1}{2}\Delta^{\text{ARE}}$ and $\Delta^{\text{sha}} = \Delta^{\text{FL}}$. The main result of this work can be stated as follows.

Theorem 4. For some leakage metric Δ^{sha} , consider a MasseySS($n, t, \mathcal{M}$) scheme over $\mathbb{F}_q$ with the δ -noisy leakage model, and leakage variable $\mathbf{L}$. Then

$$\Delta^{\text{sec}}(S; \mathbf{L}) \leq (q^{n-t+1} - 1) \cdot \eta_{\mathcal{M}}^t \leq 2^t \cdot (q^{n-t+1} - 1) \cdot \max_{\mathcal{J} \in \binom{[n]}{t}} \left(\prod_{j \in \mathcal{J}} \delta_j \right), \quad (10)$$

where $\Delta^{\text{sha}} \in \{\Delta^{\text{FL}}, \Delta^{\text{TV}}, \frac{\Delta^{\text{CDC}}}{2}, \left(\frac{\log_e 2 \cdot \text{MI}}{2}\right)^{1/2}, \frac{\Delta^{\text{ARE}}}{2}\}$, $\Delta^{\text{sec}} \in \{\Delta^{\text{FL}}, \Delta^{\text{TV}}, \frac{\Delta^{\text{CDC}}}{2}, \frac{\text{MI}}{2H(S)}, \frac{\Delta^{\text{ARE}}}{2}\}$, and $\binom{[n]}{t}$ denotes the collection of all t -element subsets of $\{1, \dots, n\}$.

For completeness, we also upper bound the pointwise TV bias, $\Delta_{\ell}^{\text{TV}}$.

Theorem 5. For a MasseySS($n, t, \mathcal{M}$) scheme over $\mathbb{F}_q$ with the δ -noisy leakage model, and some leakage outcome $\ell = (\ell_1, \dots, \ell_n) \leftarrow \mathbf{L}$; if $(q^{n-t+1} - 1)\eta_{\mathcal{M}}^t < 1$,

$$\Delta_{\ell}^{\text{TV}}(S; \mathbf{L}) \leq \frac{(q^{n-t+1} - 1) \cdot \eta_{\mathcal{M}}^t}{1 - (q^{n-t+1} - 1) \cdot \eta_{\mathcal{M}}^t}. \quad (11)$$

For an (n, t) -threshold scheme, every subset of $(t - 1)$ secret shares and the secret S are mutually independent. Therefore, a δ -noisy adversary can glean no information about the secret from less than t noisy observations. This is equivalent to the information-theoretic security of (n, t) -threshold schemes with collusion. Therefore, the bound in Theorem 4 can be interpreted as a union bound over the leakage observed by the adversary from all possible secret recovery equations in $\mathcal{C}_{\mathcal{M}}$, where leakage from each equation is upper bounded by $\eta_{\mathcal{M}}^t$.

Threshold t . For the normalized threshold $\tau = t/(n+1)$, note that $q^{n-t+1}\eta_{\mathcal{M}}^t < 1$, if and only if $\tau > 1/(1 - \log_q \eta_{\mathcal{M}})$. This provides a condition on the given scheme to be secure against leakage with some known security parameter $\eta_{\mathcal{M}}$. To characterize the rate of decay of the leaked information with increasing threshold, consider some (τ, κ) such that $q^{n-t+1}\eta_{\mathcal{M}}^t \leq \eta_{\mathcal{M}}^{\kappa t}$. This is equivalent to saying $\tau \geq \frac{1}{1 - (1 - \kappa) \log_q \eta_{\mathcal{M}}}$, which gives us the following corollary.

Corollary 3. For a MasseySS($n, t, \mathcal{M}$) scheme over $\mathbb{F}_q$, with the δ -noisy leakage model, let $\frac{t}{n+1} \geq \frac{1}{1 - (1 - \kappa) \log_q \eta_{\mathcal{M}}}$ for some $\kappa < 1$. Then

$$\Delta^{\text{sec}}(S; \mathbf{L}) \leq \eta_{\mathcal{M}}^{\kappa t}. \quad (12)$$

where $\Delta^{\text{sec}} \in \{\Delta^{\text{FL}}, \Delta^{\text{TV}}, \frac{\Delta^{\text{CDC}}}{2}, \frac{\text{MI}}{2\text{H}(S)}, \frac{\Delta^{\text{ARE}}}{2}\}$.

For bounded leakage models, Massey's scheme has been shown to be secure only when $t \geq 0.66n$ [27]. Corollary 3 shows that for any $\tau < 1$, the Massey's secret sharing scheme is secure for a sufficiently small security parameter $\eta_{\mathcal{M}}$. In particular, the expression in Theorem 4 can be rewritten as follows.

Corollary 4. For a MasseySS($n, t, \mathcal{M}$) scheme with the δ -noisy leakage model, let $t \geq \tau(n+1)$ for some $\tau < 1$. Then for the security parameter $\eta_{\mathcal{M}} \leq q^{-\gamma}$,

$$\Delta^{\text{sec}}(S; \mathbf{L}) \leq q^{-t(\gamma + 1 - \frac{1}{\tau})}, \quad (13)$$

where $\gamma > (1 - \frac{1}{\tau})$, and $\Delta^{\text{sec}} \in \{\Delta^{\text{FL}}, \Delta^{\text{TV}}, \frac{\Delta^{\text{CDC}}}{2}, \frac{\text{MI}}{2\text{H}(S)}, \frac{\Delta^{\text{ARE}}}{2}\}$.

However, to design secure schemes for small $\tau = t/(n+1)$, the security parameter $\eta_{\mathcal{M}}$ needs to be made very small. For example, to design secure schemes with $t = 0.5n$, we require $\eta_{\mathcal{M}} \leq 1/q$. This provides cryptographic engineers useful design inputs for choosing additional security measures (such as noise amplification, or wire shuffling) to protect each secret share.

Security parameter $\eta_{\mathcal{M}}$. Informally, the security parameter $\eta_{\mathcal{M}}$ can be considered to quantify the capability of the adversary in extracting *average* information from some available leakage channel. Intuitively, this capability should

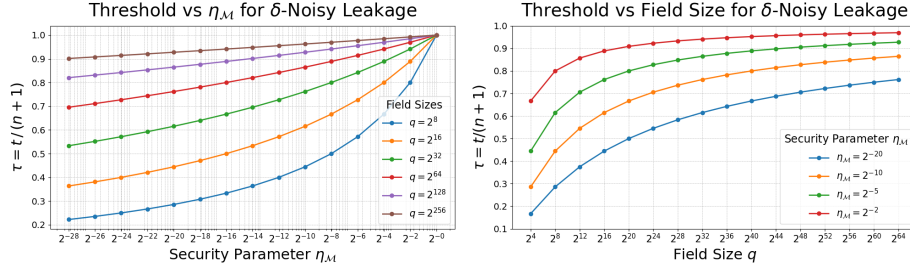


Fig. 1. For the δ -noisy leakage model over field $\mathbb{F}_q$, the security parameter $\eta_{\mathcal{M}} \leq \min\{1, 2\delta\}$. The figures represent the variation of normalized threshold $\tau = t/(n+1)$ with the security parameter $\eta_{\mathcal{M}}$ and field size q . Note that for $\eta_{\mathcal{M}} = 0$, the adversary does not leak any shares, and therefore has no information to recover the secret.

stay independent from the implementation of the underlying (n, t) -secret sharing scheme. Hence, we should expect $\eta_{\mathcal{M}}$ to remain independent of the threshold t , and the size of the underlying field $\mathbb{F}_q$. For some fixed normalized threshold $\tau = t/(n+1)$ and security parameter $\eta_{\mathcal{M}}$, Corollary 3 shows that the security guarantees worsen with the size of the field q . This is because the difficulty of reconstructing the secret from noisy observations increases with the size of the field $\mathbb{F}_q$, while the ability of the adversary does not. For instance, consider the ϵ_{RP} -random probing model over some binary extension field $\mathbb{F}_{2^\lambda}$. As shown in Proposition 2, for the same adversary, the security parameter $\eta_{\mathcal{M}} = 2\Delta^{\text{FL}} = \epsilon_{\text{RP}}$ remains constant regardless of the field size $q = 2^\lambda$. In practical implementations, such as for field size $q = 2^8$ and security parameter $\eta_{\mathcal{M}} = 0.1$, Corollary 3 yields the bound $t > 0.708n$, with security guarantees exponentially increasing as the threshold moves closer to n .

Finite Field $\mathbb{F}_q$. The bounds presented in Theorem 4, do not require any constraints on the size or characteristic of the field $\mathbb{F}_q$. In contrast to the δ -noisy leakage model, the Shamir's secret sharing scheme is known to be insecure against even 1-bit bounded leakage attacks in fields of characteristic 2. This seems to suggest that the insecurity of (n, t) -threshold schemes over fields $\mathbb{F}_{2^\lambda}$ is an artifact of the worst-case leakage functions considered in such models. Several results in leakage-resilience of Shamir's secret sharing require bounds on the size of field size. Our results show that the Shamir's scheme is provably secure for noisy leakage models, over all finite fields, at least for large thresholds.

Security for Other Leakage Models. Theorem 4 allows us to evaluate the security guarantees for any noisy leakage framework which can be subsumed under the δ -noisy leakage model. For example, consider the ϵ_{RP} -random probing model (Section 4.2), where the i^{th} leakage function L_i is an ϵ_i -RP leakage of U_i . From Proposition 2, recall that $2\Delta^{\text{FL}}(U_i, L_i) = \epsilon_i$. Plugging these in Theorem 4, we get the following corollary.

Corollary 5. Consider a MasseySS($n, t, \mathcal{M}$) scheme over $\mathbb{F}_q$ with the ϵ_{RP} -random probing model, and leakage variable $\mathbf{L}$,

$$\Delta^{\text{sec}}(S; \mathbf{L}) \leq (q^{n-t+1} - 1) \cdot \max_{\mathcal{J} \in \binom{[n]}{t}} \left(\prod_{j \in \mathcal{J}} \epsilon_j \right), \quad (14)$$

where $\Delta^{\text{sec}} \in \{\Delta^{\text{FL}}, \Delta^{\text{TV}}, \frac{\Delta^{\text{CDC}}}{2}, \frac{\text{MI}}{2\text{H}(S)}, \frac{\Delta^{\text{ARE}}}{2}\}$, and $\binom{[n]}{t}$ denotes the collection of all t -element subsets of $\{1, \dots, n\}$.

5.3 Security of (n, n) -Threshold Schemes

Several recent works have investigated the security of masking schemes with δ -noisy leakage [3, 4, 9, 16, 17, 18, 20, 24, 29, 39, 43, 45]. For masked encodings, i.e. (n, n) -threshold schemes over arbitrary fields with δ -noisy leakage, the previous best known result is due to Dziembowski, Faust, and Skórski (TCC'16) [17]. The authors first reduce the security analysis to uniform secrets and consequently independent secret shares, and then use the XOR lemma. However, this indirect method leads to a loss in security guarantees for small δ . Theorem 4 allows us to bound the security of masked encodings without relying upon such reductions.

Corollary 6. For some leakage metric Δ^{sha} , consider an (n, n) -threshold secret sharing scheme over $\mathbb{F}_q$ with the δ -noisy leakage model, and some leakage $\mathbf{L}$,

$$\Delta^{\text{sec}}(S; \mathbf{L}) \leq (q - 1) \cdot \eta_{\mathcal{M}}^n \leq 2^n \cdot (q - 1) \cdot \left(\prod_{i=1}^n \delta_i \right). \quad (15)$$

where $\Delta^{\text{sha}} \in \{\Delta^{\text{FL}}, \Delta^{\text{TV}}, \frac{\Delta^{\text{CDC}}}{2}, \left(\frac{\log_e 2 \cdot \text{MI}}{2}\right)^{1/2}, \frac{\Delta^{\text{ARE}}}{2}\}$, $\Delta^{\text{sec}} \in \{\Delta^{\text{FL}}, \Delta^{\text{TV}}, \frac{\Delta^{\text{CDC}}}{2}, \frac{\text{MI}}{2\text{H}(S)}, \frac{\Delta^{\text{ARE}}}{2}\}$, and $\binom{[n]}{t}$ denotes the collection of all t -element subsets of $\{1, \dots, n\}$.

For fields of characteristic 2, Ito et al. [24] used similar Fourier analytical methods to bound the attack success rate. In Corollary 7, we bound several other leakage metrics being considered in this work and show that the leakage is $\mathcal{O}(2 \cdot q^{1/n} \cdot \delta)^n$. Therefore, for binary fields $\mathbb{F}_{2^\lambda}$, even a small number of shares $n = c\lambda$, gives us exponentially strong security guarantees.

Corollary 7. For some binary extension field $\mathbb{F}_{2^\lambda}$, consider an (n, n) -threshold secret sharing scheme with the δ -noisy leakage model, where $n \geq c\lambda$. Then

$$\Delta^{\text{sec}}(S; \mathbf{L}) < (2^{1/c} \cdot \eta_{\mathcal{M}})^{c\lambda} \leq (2^{1+1/c} \cdot \max_{j \in [n]} \Delta^{\text{TV}}(U_j; L_j))^{c\lambda}. \quad (16)$$

where $\Delta^{\text{sec}} \in \{\Delta^{\text{FL}}, \Delta^{\text{TV}}, \frac{\Delta^{\text{CDC}}}{2}, \frac{\text{MI}}{2\text{H}(S)}, \frac{\Delta^{\text{ARE}}}{2}\}$, and $[n] = \{1, \dots, n\}$.

For prime fields $\mathbb{F}_p$, Dziembowski et al. [17] show that the minimum required number of shares to guarantee security is $n_{\min} > 2 \ln p \cdot (1 - 1/p - \delta)^{-4}$. However, even for small δ , $n_{\min}$ remains large and does not improve much as δ decreases.

Therefore, this could not have been used by designers to claim any practically relevant formal security guarantees. The results in Corollary 7 show that when the number of shares $n \geq -\ln p / \ln 2\delta$, the overall leakage is less than 1; thereby making the implementation secure. This result is particularly useful in designing secure cryptographic schemes operating over large fields with concrete security guarantees. e.g., the AES or the recently standardized post-quantum schemes. In cryptographic applications, typically $n = 2, 3, 4$, and $q = 2^8$ for AES, $\approx 2^{12}$ for Kyber, $\approx 2^{23}$ for Dilithium, and $\approx 2^{49}$ for the signature scheme Raccoon. In Figure 2, for these field sizes, we plot the minimum number of shares required that guarantee the security of (n, n) -threshold schemes for some leakage parameter δ . Additionally, for field sizes $q = 2^8, 2^{12}$, we demonstrate how the overall leakage changes with the security parameter $\eta_{\mathcal{M}}$ for $n = 2, 3, 5, 10$.

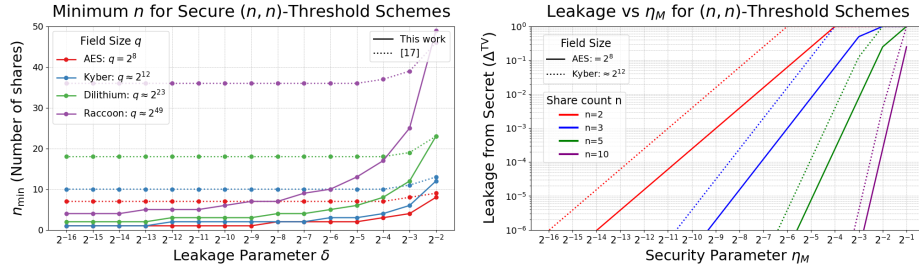


Fig. 2. The figures illustrate the security trade-offs for (n, n) -threshold schemes over finite fields. Left: comparison between the bound established in this work and that of Dziembowski et al. [17], showing the minimum share count $n_{\min} = \lceil -\ln q / \ln 2\delta \rceil$ required to guarantee security against δ -noisy leakage. The value of $n_{\min}$ is plotted as a function of the leakage parameter δ for field sizes $q \in \{2^8, \approx 2^{12}, \approx 2^{23}, \approx 2^{49}\}$. Right: total variation leakage Δ^{TV} as a function of $\eta_{\mathcal{M}}$ for $q \in \{2^8, \approx 2^{12}\}$ and $n \in \{2, 3, 5, 10\}$.

6 Formal Security Proofs

In this section, we prove the main results of this work presented in Section 5. Consider an adversary who wishes to reconstruct the secret $s \leftarrow S$ from some leakage outcome $\ell \leftarrow \mathbf{L} = (L_1, \dots, L_n)$ for some realization of the underlying secret sharing scheme with the δ -noisy leakage (Section 4). We assume that the adversary knows the codebook $\mathcal{C}_{\mathcal{M}}$, and therefore the linear relations satisfied by the secret shares, which correspond to the secret recovery equations. These relations induce *dependencies* among the secret shares constraining their joint independence, which then (unfortunately) propagates to the leakage $\mathbf{L}$. In the δ -noisy leakage model, we assume that the leakage channels $L_i|U_i$ are mutually independent. Assume for instance, if the leakage observed from individual secret shares were also to be mutually independent, i.e. $\mathbb{P}(\mathbf{L} = \ell) = \prod_{i=1}^n \mathbb{P}(L_i = \ell_i)$; then the adversary would gain no additional information about the secret from the algebraic structure of the scheme. The optimal strategy for such an

adversary is to guess each secret share individually and independently from its leakage. Intuitively, analyzing such an adversary may offer us an indirect path to bounding the overall leakage from the secret. To formalize this intuition, we define the functions Λ and Λ' as follows.

Definition 17. *Given secret $s \leftarrow S$, and some leakage outcome $\ell \leftarrow \mathbf{L}$, define functions $\Lambda : \mathbb{F}_q \otimes \mathcal{L} \rightarrow \mathbb{R}$ and $\Lambda' : \mathbb{F}_q \otimes \mathcal{L} \rightarrow \mathbb{R}$ as*

$$\Lambda(s, \ell) = \frac{\mathbb{P}(S = s, \mathbf{L} = \ell)}{\mathbb{P}(S = s)\mathbb{P}(\mathbf{L} = \ell)}, \text{ and } \Lambda'(s, \ell) = \frac{\mathbb{P}(S = s, \mathbf{L} = \ell)}{\mathbb{P}(S = s) \prod_{i=1}^n \mathbb{P}(L_i = \ell_i)}.$$

The function $\Lambda(s, \ell)$ appears naturally in all the metrics considered in this work (Section 3), and is the central object of study in this section:

- $2\Delta_{\ell}^{\text{TV}}(S; \mathbf{L}) = \mathbb{E}_S[|\Lambda - 1|]$, and $2\Delta^{\text{TV}}(S; \mathbf{L}) = \mathbb{E}_{\mathbf{L}}\mathbb{E}_S[|\Lambda - 1|]$,
- $\text{MI}(S; \mathbf{L}) = \mathbb{E}_{(S, \mathbf{L})}[\log_2(\Lambda)]$,
- $\Delta^{\text{CDC}}(S; \mathbf{L}) = \mathbb{E}_{\mathbf{L}}[\max_s |\Lambda - 1|]$, and $\Delta^{\text{ARE}}(S; \mathbf{L}) = \mathbb{E}_{\mathbf{L}}[\max_s |\Lambda - 1|]$.

Informally, for a given secret s , the function Λ' can be understood to be measuring the impact of the structure of the underlying scheme on the independence of leaked secret shares. In Lemma 4, we upper bound $\max_{s \in \mathbb{F}_q} |\Lambda' - 1|$, providing the bridge between algebraic properties of the underlying scheme and the analytic bounds required for our main theorems. Using the triangle inequality,

$$|\Lambda(s, \ell) - 1| \leq \frac{\Lambda(s, \ell)}{\Lambda'(s, \ell)} \left(|\Lambda'(s, \ell) - 1| + \left| \frac{\Lambda'(s, \ell)}{\Lambda(s, \ell)} - 1 \right| \right). \quad (17)$$

The following proposition will then make it possible to bound the required $\max_{s \in \mathbb{F}_q} |\Lambda(s, \ell) - 1|$ in Section 6.2.

Proposition 4. *Given the secret $s \leftarrow S$, and some leakage outcome $\ell \leftarrow \mathbf{L}$,*

$$\left| \frac{\Lambda'(s, \ell)}{\Lambda(s, \ell)} - 1 \right| \leq \max_{s \in \mathbb{F}_q} |(\Lambda'(s, \ell) - 1)|.$$

Proof. To prove this result, observe that

$$\left| \frac{\Lambda'}{\Lambda} - 1 \right| = \left| \frac{\mathbb{P}(\mathbf{L} = \ell)}{\prod_{i=1}^n \mathbb{P}(L_i = \ell_i)} - 1 \right| = |\mathbb{E}_S[(\Lambda'(S, \ell) - 1)]| \leq \mathbb{E}_S[|(\Lambda'(S, \ell) - 1)|].$$

Now using the triangle inequality in (17), we obtain the following.

Corollary 8. *Given the secret $s \leftarrow S$, and some leakage outcome $\ell \leftarrow \mathbf{L}$,*

$$\max_{s \in \mathbb{F}_q} |\Lambda(s, \ell) - 1| \leq 2 \cdot \frac{\Lambda(s, \ell)}{\Lambda'(s, \ell)} \cdot \left(\max_{s \in \mathbb{F}_q} |\Lambda'(s, \ell) - 1| \right).$$

To prove the bound on $|\Lambda'(s, \ell) - 1|$ in Lemma 4, we will first need to rewrite the expression for $\mathbb{P}(\mathbf{L} = \ell | S = s)$ in such a way that we can exploit the independence of leakage channels. For this purpose, we marginalize this expression

over $\mathbf{U}$, and then using the Bayes' rule decompose Λ' into sums over conditional marginal probabilities of the leakage channels. Then a straightforward application of the Poisson summation formula (Theorem 2) yields the stated result. For some leakage ℓ , we introduce the following notation for the conditional marginal probabilities and their product.

Definition 18. *Given some leakage outcome $\ell \leftarrow \mathbf{L}$, for $1 \leq i \leq n$, define functions $\Psi_{\ell_i} : \mathbb{F}_q \rightarrow \mathbb{C}$ such that $\Psi_{\ell_i}(\alpha) = \mathbb{P}(U_i = \alpha | L_i = \ell_i)$, and let $\Psi_\ell : \mathbb{F}_q^n \rightarrow \mathbb{C}$ be the function defined as $\Psi_\ell(\boldsymbol{\alpha} = (\alpha_1, \dots, \alpha_n)) = \prod_{i=1}^n \Psi_{\ell_i}(\alpha_i)$.*

Lemma 4. *For a MasseySS($n, t, \mathcal{M}$) scheme over $\mathbb{F}_q$, with secret $s \leftarrow S$, under the δ -noisy leakage model, then*

$$\max_{s \in \mathbb{F}_q} |\Lambda'(s, \ell) - 1| \leq \sum_{\mathbf{w} \in \mathcal{C}_{\mathcal{M}}^\times} |\hat{\Psi}_\ell(\mathbf{w})|. \quad (18)$$

Proof. For some secret s , the corresponding secret shares' vector $\mathbf{u}$ lies in some coset of the dual code of $\mathcal{C}_{\mathcal{M}}$, i.e. $\mathbf{u} \in \mathbf{v} + \mathcal{C}_{\mathcal{M}}^\perp$ for some $\mathbf{v} \in \mathbb{F}_q^n$. To prove the bound in (24), we will need to rewrite the expression for $\mathbb{P}(\mathbf{L} = \ell | S = s)$ in such a way that we can reduce it to analyzing individual leakage channels, and then subsequently use the bounds on leakage from each secret share individually.

$$\begin{aligned} \mathbb{P}(\mathbf{L} = \ell | S = s) &= \frac{\mathbb{P}(\mathbf{L} = \ell, S = s)}{\mathbb{P}(S = s)} \\ &= \frac{\sum_{\mathbf{u}: S=s} p_{\mathbf{U}}(\mathbf{u}) \mathbb{P}(\mathbf{L} = \ell | \mathbf{U} = \mathbf{u})}{\sum_{\mathbf{u}: S=s} p_{\mathbf{U}}(\mathbf{u})} \end{aligned} \quad (19)$$

$$= q^{k-n} \cdot \left(\sum_{\mathbf{u}: S=s} \mathbb{P}(\mathbf{L} = \ell | \mathbf{U} = \mathbf{u}) \right) \quad (20)$$

$$= q^{k-n} \cdot \left(\sum_{\mathbf{u}: S=s} \prod_{i=1}^n \mathbb{P}(L_i = \ell_i | U_i = u_i) \right) \quad (21)$$

$$= q^k \cdot \prod_{i=1}^n \mathbb{P}(L_i = \ell_i) \cdot \left(\sum_{\mathbf{u}: S=s} \prod_{i=1}^n \mathbb{P}(U_i = u_i | L_i = \ell_i) \right), \quad (22)$$

where (20) follows from (19) by Proposition 1, and (21) follows from (20) from the independence of leakage channels $L_i | U_i$. Since the secret shares are uniformly distributed, (22) follows from (21) as: $\mathbb{P}(L_i | U_i) = \mathbb{P}(L_i) \cdot \mathbb{P}(U_i | L_i) / \mathbb{P}(U_i) = q \cdot \mathbb{P}(L_i) \cdot \mathbb{P}(U_i | L_i)$. Finally, the expression in (22) can be rewritten as $\mathbb{P}(\mathbf{L} = \ell | S = s) = q^k \cdot \prod_{i=1}^n \mathbb{P}(L_i = \ell_i) \cdot \left(\sum_{\mathbf{u} \in \mathbf{v} + \mathcal{C}_{\mathcal{M}}^\perp} \Psi_\ell(\mathbf{u}) \right)$, for some $\mathbf{v} \in \mathbb{F}_q^n$. Using the Poisson Summation formula from Theorem 2,

$$\Lambda'(s, \ell) = \frac{\mathbb{P}(\mathbf{L} = \ell | S = s)}{\prod_{i=1}^n \mathbb{P}(L_i = \ell_i)} = q^k \cdot \sum_{\mathbf{u} \in \mathbf{v} + \mathcal{C}_{\mathcal{M}}^\perp} \Psi_\ell(\mathbf{u}) = \sum_{\mathbf{w} \in \mathcal{C}_{\mathcal{M}}} \hat{\Psi}_\ell(\mathbf{w}) \chi(-\langle \mathbf{v}, \mathbf{w} \rangle). \quad (23)$$

Since $\hat{\Psi}_\ell(\mathbf{0}) \chi(0) = 1$, and $|\chi(-\langle \mathbf{v}, \mathbf{w} \rangle)| = 1$ for all $\mathbf{w} \in \mathcal{C}_{\mathcal{M}}$, we get $|\Lambda'(s, \ell) - 1| \leq \left| \sum_{\mathbf{w} \in \mathcal{C}_{\mathcal{M}}^\times} \hat{\Psi}_\ell(\mathbf{w}) \chi(-\langle \mathbf{v}, \mathbf{w} \rangle) \right| \leq \sum_{\mathbf{w} \in \mathcal{C}_{\mathcal{M}}^\times} |\hat{\Psi}_\ell(\mathbf{w})|$.

6.1 Proof of Theorem 5

So far we have only used the fact that the secret shares' vectors $\mathbf{u} \in \mathcal{U}$ form an affine subspace of $\mathbb{F}_q^n$. Recall that the threshold t of the underlying scheme is such that, for any $\mathbf{w} \in \mathcal{C}_{\mathcal{M}}^\times$, we have $\text{wt}(\mathbf{w}) \geq t$. In the following corollary, we will demonstrate how to use this fact to bound $|\widehat{\Psi}_{\ell}(\mathbf{w})|$. Arguments similar to the proof of Proposition 5, will be used again in bounding $\Delta^{\text{ARE}}(S; \mathbf{L})$.

Proposition 5. *For a MasseySS($n, t, \mathcal{M}$) scheme over $\mathbb{F}_q$, with secret $s \leftarrow S$, under the δ -noisy leakage model, then*

$$|\Lambda'(s, \ell) - 1| \leq 2^t \cdot (q^{n-t+1} - 1) \cdot \left(\max_{\mathcal{J} \in \binom{[n]}{t}} \prod_{j \in \mathcal{J}} \Delta_{\ell_j}^{\text{FL}}(U_j, L_j) \right), \quad (24)$$

where $\binom{[n]}{t}$ denotes the collection of all t -element subsets of $\{1, \dots, n\}$.

Proof. For any $\mathbf{w}$, we have that $|\widehat{\Psi}_{\ell}(\mathbf{w})| = \prod_{i=1}^n |\widehat{\Psi}_{\ell_i}(w_i)|$, where $|\widehat{\Psi}_{\ell_i}(0)| = 1$. Since each secret share U_i is uniformly distributed, for $w_i \neq 0$, $|\widehat{\Psi}_{\ell_i}(w_i)| \leq \text{bias}(U_i | L_i = \ell_i) = 2 \Delta_{\ell_i}^{\text{FL}}(U_i, L_i)$. Recall from Lemma 1 that $\mathcal{C}_{\mathcal{M}}$ is an $[n, k, t]$ code, i.e. its minimum weight is the threshold t . For some $\mathbf{w} \in \mathcal{C}_{\mathcal{M}}^\times$, let $\mathcal{J} \subseteq \text{supp}(\mathbf{w})$ be a subset of size t in the support of $\mathbf{w}$. Then $\mathcal{J} \in \binom{[n]}{t}$, and

$$|\widehat{\Psi}_{\ell}(\mathbf{w})| \leq \prod_{j \in \mathcal{J}} |\widehat{\Psi}_{\ell_j}(w_j)| \leq 2^t \cdot \prod_{j \in \mathcal{J}} \Delta_{\ell_j}^{\text{FL}}(U_j, L_j) \leq 2^t \cdot \max_{\mathcal{J} \in \binom{[n]}{t}} \prod_{j \in \mathcal{J}} \Delta_{\ell_j}^{\text{FL}}(U_j, L_j).$$

Using the Singleton bound, we see that $|\mathcal{C}_{\mathcal{M}}| \leq q^{n-t+1}$. Therefore,

$$|\Lambda'(s, \ell) - 1| \leq \sum_{\mathbf{w} \in \mathcal{C}_{\mathcal{M}}^\times} |\widehat{\Psi}_{\ell}(\mathbf{w})| \leq 2^t \cdot (q^{n-t+1} - 1) \cdot \left(\max_{\mathcal{J} \in \binom{[n]}{t}} \prod_{j \in \mathcal{J}} \Delta_{\ell_j}^{\text{FL}}(U_j, L_j) \right).$$

Proof of Theorem 5. To prove bounds for the pointwise TV Bias in Theorem 5, recall that $2\Delta_{\ell}^{\text{TV}}(S; \mathbf{L}) = \mathbb{E}_S [|\Lambda(s, \ell) - 1|] \leq \max_{s \in \mathbb{F}_q} |\Lambda(s, \ell) - 1|$. From Proposition 4 and Corollary 8,

$$\Delta_{\ell}^{\text{TV}}(S; \mathbf{L}) \leq \frac{\Lambda(s, \ell)}{\Lambda'(s, \ell)} \cdot \max_{s \in \mathbb{F}_q} |\Lambda'(s, \ell) - 1| \leq \frac{2 \cdot \max_{s \in \mathbb{F}_q} |\Lambda'(s, \ell) - 1|}{1 - \max_{s \in \mathbb{F}_q} |\Lambda'(s, \ell) - 1|}.$$

Finally, using the bounds from Proposition 5 yields Theorem 5.

6.2 Proof of Theorem 4

To finally prove the main result of this work, recall that it enough to prove Theorem 4 for $\Delta^{\text{sec}} = \frac{1}{2} \Delta^{\text{ARE}}$. From Definition 11, recall that $\Delta^{\text{ARE}}(S; \mathbf{L}) = \mathbb{E}_{\mathbf{L}} [\max_{s \in \mathbb{F}_q} |\Lambda(s, \mathbf{L}) - 1|]$. From Corollary 8, observe

$$\frac{1}{2} \Delta^{\text{ARE}}(S; \mathbf{L}) \leq \mathbb{E}_{\mathbf{L}} \left[\frac{\Lambda(s, \ell)}{\Lambda'(s, \ell)} \max_{s \in \mathbb{F}_q} |\Lambda'(s, \ell) - 1| \right].$$

Now using Lemma 4,

$$\begin{aligned}
\mathbb{E}_{\mathbf{L}} \left[\frac{\Lambda(s, \ell)}{\Lambda'(s, \ell)} \max_{s \in \mathbb{F}_q} |\Lambda'(s, \ell) - 1| \right] &= \sum_{\ell \in \mathcal{L}} \left(\prod_{i=1}^n \mathbb{P}(L_i = \ell_i) \right) \cdot \max_{s \in \mathbb{F}_q} |\Lambda'(s, \ell) - 1| \\
&\leq \sum_{\ell \in \mathcal{L}} \left(\prod_{i=1}^n \mathbb{P}(L_i = \ell_i) \right) \cdot \left(\sum_{\mathbf{w} \in \mathcal{C}_{\mathcal{M}}^{\times}} |\hat{\Psi}_{\ell}(\mathbf{w})| \right) \\
&\leq \sum_{\mathbf{w} \in \mathcal{C}_{\mathcal{M}}^{\times}} \sum_{\ell \in \mathcal{L}} \prod_{i=1}^n \mathbb{P}(L_i = \ell_i) \cdot |\hat{\Psi}_{\ell_i}(w_i)| \\
&\leq \sum_{\mathbf{w} \in \mathcal{C}_{\mathcal{M}}^{\times}} \prod_{i=1}^n \left(\sum_{\ell_i \in \mathcal{L}_i} \mathbb{P}(L_i = \ell_i) \cdot |\hat{\Psi}_{\ell_i}(w_i)| \right) \\
&\leq \sum_{\mathbf{w} \in \mathcal{C}_{\mathcal{M}}^{\times}} \prod_{i=1}^n \mathbb{E}_{L_i} [|\hat{\Psi}_{\ell_i}(w_i)|].
\end{aligned}$$

Proceeding similarly to the proof of Proposition 5, for some $\mathbf{w} \in \mathcal{C}_{\mathcal{M}}^{\times}$, let $\mathcal{J} \subseteq \text{supp}(\mathbf{w})$ be a subset of size t in the support of $\mathbf{w}$. Then $\mathcal{J} \in \binom{[n]}{t}$, and for $j \in \mathcal{J}$, we have $\mathbb{E}_{L_j} [|\hat{\Psi}_{\ell_j}(\mathbf{w})|] \leq \mathbb{E}_{L_j} [2\Delta_{\ell_j}^{\text{FL}}(U_j, L_j)] \leq 2\Delta^{\text{FL}}(U_j, L_j)$. Therefore

$$\prod_{i=1}^n \mathbb{E}_{L_i} [|\hat{\Psi}_{\ell_i}(\mathbf{w})|] \leq 2^t \cdot \prod_{j \in \mathcal{J}} \Delta^{\text{FL}}(U_j, L_j) \leq 2^t \cdot \max_{\mathcal{J} \in \binom{[n]}{t}} \prod_{j \in \mathcal{J}} \Delta^{\text{FL}}(U_j, L_j).$$

Using $|\mathcal{C}_{\mathcal{M}}^{\times}| \leq (q^{n-t+1} - 1)$ gives us the required inequality

$$\begin{aligned}
\frac{1}{2} \Delta^{\text{ARE}}(S; \mathbf{L}) &\leq \mathbb{E}_{\mathbf{L}} \left[\frac{\Lambda(s, \ell)}{\Lambda'(s, \ell)} \max_{s \in \mathbb{F}_q} |\Lambda'(s, \ell) - 1| \right] \\
&\leq 2^t \cdot (q^{n-t+1} - 1) \cdot \left(\max_{\mathcal{J} \in \binom{[n]}{t}} \prod_{j \in \mathcal{J}} \Delta^{\text{FL}}(U_j, L_j) \right).
\end{aligned}$$

7 Conclusion

To examine the leakage resilience of secret sharing, we study the security of linear code-based (n, t) -threshold schemes with δ -noisy side-channel leakage. In this leakage model, each secret share is independently leaked to an adversary with some leakage parameter δ . For (n, t) -threshold schemes, we show that the security guarantees improve exponentially with t for the following leakage metrics: statistical bias (Δ^{TV}), mutual information (MI), complementary Doeblin coefficient (Δ^{CDC}), and average relative error (Δ^{ARE}). Our results are characterized by the security parameter $\eta_{\mathcal{M}}$, which is determined by the geometric mean of the worst t Fourier biases of the δ -noisy leakage channels. The security

parameter is such that $\eta_{\mathcal{M}} \leq \min\{1, 2 \cdot \max \delta_i\}$. In particular, for the normalized threshold $\tau = t/(n+1)$, we show that if $\eta_{\mathcal{M}} \leq q^{-\gamma}$ for some $\gamma > 1/\tau - 1$, then the statistical bias of the leakage is bounded by $\mathcal{O}(q^{-t(\gamma+1-1/\tau)})$, where $\mathbb{F}_q$ is the underlying field. This implies that linear secret sharing schemes with $t \geq \tau n$ are secure for a sufficiently small security parameter $\eta_{\mathcal{M}}$. Consequently, for (n, n) -threshold schemes over arbitrary prime fields, we improve the previous best known bounds by Dziembowski *et al.* for small leakage parameter δ . Our results for the first time provide practical security guarantees, which we hope are relevant to designing secure implementations of secret sharing schemes.

This naturally leads to new questions. Investigating the ultimate tightness of our bounds remains an important open problem, particularly by addressing the dependence on the code size $|\mathcal{C}_{\mathcal{M}}| \leq q^{n-t+1}$ that arises from the union bound interpretation of the results in Theorem 4. We believe that this could be improved with a more careful analysis or by extending the techniques in this work, analogous to recent advancements in the bounded leakage model. Although the statistical nature of our techniques do not present much intuition about devising optimal strategies for adversaries, it gives defenders a robust, quantitative measure of security to design against, regardless of the adversary's specific strategy. We leave it as an important open question to apply our results to some of the existing implementations of secret sharing schemes, such as in multi-party computation protocols and masked encodings. Moreover, the successful adaptation of the Fourier-analytic framework suggests its potential applicability beyond secret sharing. Investigating the noisy leakage resilience of other foundational primitives presents a promising direction for future inquiry.

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Bibliography

- [1] Adams, D.Q., Maji, H.K., Nguyen, H.H., Nguyen, M.L., Paskin-Cherniavsky, A., Suad, T., Wang, M.: Lower bounds for leakage-resilient secret-sharing schemes against probing attacks. In: 2021 IEEE International Symposium on Information Theory (ISIT). pp. 976–981. IEEE (2021)
- [2] Béguinot, J., Cheng, W., Guilley, S., Liu, Y., Masure, L., Rioul, O., Standardaert, F.X.: Removing the field size loss from duc et al.’s conjectured bound for masked encodings. In: International Workshop on Constructive Side-Channel Analysis and Secure Design. pp. 86–104. Springer (2023)
- [3] Béguinot, J., Cheng, W., Guilley, S., Rioul, O.: Formal security proofs via Doeblin coefficients: optimal side-channel factorization from noisy leakage to random probing. In: Annual International Cryptology Conference. pp. 389–426. Springer (2024)
- [4] Béguinot, J., Liu, Y., Rioul, O., Cheng, W., Guilley, S.: Maximal leakage of masked implementations using Mrs. Gerber’s lemma for min-entropy. In: 2023 IEEE International Symposium on Information Theory (ISIT). pp. 654–659. IEEE (2023)
- [5] Beimel, A.: Secret-sharing schemes: A survey. In: International conference on coding and cryptology. pp. 11–46. Springer (2011)
- [6] Benhamouda, F., Degwekar, A., Ishai, Y., Rabin, T.: On the Local Leakage Resilience of Linear Secret Sharing Schemes: 38th Annual International Cryptology Conference, Santa Barbara, CA, USA, August 19–23, 2018, Proceedings, Part I, pp. 531–561 (07 2018)
- [7] Benhamouda, F., Degwekar, A., Ishai, Y., Rabin, T.: On the local leakage resilience of linear secret sharing schemes. *Journal of Cryptology* **34**, 1–65 (2021)
- [8] Blakley, G.R.: Safeguarding cryptographic keys. In: Managing Requirements Knowledge, International Workshop on. pp. 313–313. IEEE Computer Society (1979)
- [9] Brian, G., Dziembowski, S., Faust, S.: From random probing to noisy leakages without field-size dependence. In: Annual International Conference on the Theory and Applications of Cryptographic Techniques. pp. 345–374. Springer (2024)
- [10] Brian, G., Faonio, A., Obremski, M., Ribeiro, J., Simkin, M., Skórski, M., Venturi, D.: The mother of all leakages: how to simulate noisy leakages via bounded leakage (almost) for free. *IEEE Transactions on Information Theory* **68**(12), 8197–8227 (2022)
- [11] Brickell, E.F.: Some ideal secret sharing schemes. In: Workshop on the Theory and Application of of Cryptographic Techniques. pp. 468–475. Springer (1989)
- [12] Cheng, W., Liu, Y., Guilley, S., Rioul, O.: Attacking masked cryptographic implementations: Information-theoretic bounds. In: 2022 IEEE International Symposium on Information Theory (ISIT). pp. 654–659. IEEE (2022)

- [13] Cramer, R., Daza, V., Gracia, I., Urroz, J.J., Leander, G., Martí-Farré, J., Padró, C.: On codes, matroids, and secure multiparty computation from linear secret-sharing schemes. *IEEE Transactions on Information Theory* **54**(6), 2644–2657 (2008)
- [14] Duc, A., Dziembowski, S., Faust, S.: Unifying leakage models: from probing attacks to noisy leakage. In: *Advances in Cryptology–EUROCRYPT 2014: 33rd Annual International Conference on the Theory and Applications of Cryptographic Techniques*, Copenhagen, Denmark, May 11–15, 2014. *Proceedings* 33. pp. 423–440. Springer (2014)
- [15] Duc, A., Faust, S., Standaert, F.X.: Making masking security proofs concrete: or how to evaluate the security of any leaking device. In: *Advances in Cryptology–EUROCRYPT 2015: 34th Annual International Conference on the Theory and Applications of Cryptographic Techniques*, Sofia, Bulgaria, April 26–30, 2015, *Proceedings, Part I* 34. pp. 401–429. Springer (2015)
- [16] Dziembowski, S., Faust, S., Skorski, M.: Noisy leakage revisited. In: *Annual International Conference on the Theory and Applications of Cryptographic Techniques*. pp. 159–188. Springer (2015)
- [17] Dziembowski, S., Faust, S., Skórski, M.: Optimal amplification of noisy leakages. In: *Theory of Cryptography: 13th International Conference, TCC 2016-A*, Tel Aviv, Israel, January 10–13, 2016, *Proceedings, Part II* 13. pp. 291–318. Springer (2016)
- [18] Faust, S., Masure, L., Micheli, E., Orlt, M., Standaert, F.X.: Connecting leakage-resilient secret sharing to practice: Scaling trends and physical dependencies of prime field masking. In: *Annual International Conference on the Theory and Applications of Cryptographic Techniques*. pp. 316–344. Springer (2024)
- [19] Faust, S., Rabin, T., Reyzin, L., Tromer, E., Vaikuntanathan, V.: Protecting circuits from leakage: the computationally-bounded and noisy cases. In: *Advances in Cryptology–EUROCRYPT 2010: 29th Annual International Conference on the Theory and Applications of Cryptographic Techniques*, French Riviera, May 30–June 3, 2010. *Proceedings* 29. pp. 135–156. Springer (2010)
- [20] Gupta, U., MahdaviFar, H.: Bounds on the statistical leakage-resilience of Shamir’s secret sharing. In: *2024 IEEE International Symposium on Information Theory (ISIT)*. pp. 184–189 (2024). <https://doi.org/10.1109/ISIT57864.2024.10619359>
- [21] Guruswami, V., Wootters, M.: Repairing Reed-Solomon codes. *IEEE Transactions on Information Theory* **63**(9), 5684–5698 (2017). <https://doi.org/10.1109/TIT.2017.2702660>
- [22] Hoffmann, C., Simkin, M.: Stronger lower bounds for leakage-resilient secret sharing. In: *International Conference on Cryptology and Information Security in Latin America*. pp. 215–228. Springer (2023)
- [23] Ishai, Y., Sahai, A., Wagner, D.: Private circuits: Securing hardware against probing attacks. In: *Advances in Cryptology–CRYPTO 2003: 23rd Annual International Cryptology Conference*, Santa Barbara, California, USA, August 17–21, 2003. *Proceedings* 23. pp. 463–481. Springer (2003)

- [24] Ito, A., Ueno, R., Homma, N.: On the success rate of side-channel attacks on masked implementations: information-theoretical bounds and their practical usage. In: Proceedings of the 2022 ACM SIGSAC Conference on Computer and Communications Security. pp. 1521–1535 (2022)
- [25] Jahandideh, V., Mennink, B., Batina, L.: A decomposition approach for evaluating security of masking. In: International Conference on the Theory and Application of Cryptology and Information Security. pp. 580–612. Springer (2025)
- [26] Kalai, Y.T., Reyzin, L.: A survey of leakage-resilient cryptography. In: Providing Sound Foundations for Cryptography: On the Work of Shafi Goldwasser and Silvio Micali, pp. 727–794 (2019)
- [27] Kasser, D.: An improvement upon the bounds for the local leakage resilience of Shamir’s secret sharing scheme. In: Theory of Cryptography Conference. pp. 395–422. Springer (2025)
- [28] Klein, O., Komargodski, I.: New bounds on the local leakage resilience of Shamir’s secret sharing scheme. In: Annual International Cryptology Conference. pp. 139–170. Springer (2023)
- [29] Liu, Y., Béguinot, J., Cheng, W., Guilley, S., Masure, L., Rioul, O., Standardert, F.X.: Improved alpha-information bounds for higher-order masked cryptographic implementations. In: 2023 IEEE Information Theory Workshop (ITW). pp. 81–86. IEEE (2023)
- [30] Maji, H.K., Nguyen, H.H., Paskin-Cherniavsky, A., Suad, T., Wang, M.: Leakage-resilience of the Shamir secret-sharing scheme against physical-bit leakages. In: Annual International Conference on the Theory and Applications of Cryptographic Techniques. pp. 344–374. Springer (2021)
- [31] Maji, H.K., Nguyen, H.H., Paskin-Cherniavsky, A., Suad, T., Wang, M., Ye, X., Yu, A.: Leakage-resilient linear secret-sharing against arbitrary bounded-size leakage family. In: Theory of Cryptography Conference. pp. 355–383. Springer (2022)
- [32] Maji, H.K., Nguyen, H.H., Paskin-Cherniavsky, A., Suad, T., Wang, M., Ye, X., Yu, A.: Tight estimate of the local leakage resilience of the additive secret-sharing scheme & its consequences. In: 3rd Conference on Information-Theoretic Cryptography (ITC 2022). Schloss Dagstuhl-Leibniz-Zentrum für Informatik (2022)
- [33] Maji, H.K., Nguyen, H.H., Paskin-Cherniavsky, A., Wang, M.: Improved bound on the local leakage-resilience of Shamir’s secret sharing. In: 2022 IEEE International Symposium on Information Theory (ISIT). pp. 2678–2683. IEEE (2022)
- [34] Maji, H.K., Nguyen, H.H., Paskin-Cherniavsky, A., Ye, X.: Constructing leakage-resilient Shamir’s secret sharing: Over composite order fields. In: Annual International Conference on the Theory and Applications of Cryptographic Techniques. pp. 286–315. Springer (2024)
- [35] Maji, H.K., Paskin-Cherniavsky, A., Suad, T., Wang, M.: Constructing locally leakage-resilient linear secret-sharing schemes. In: Annual International Cryptology Conference. pp. 779–808. Springer (2021)

- [36] Mangard, S., Oswald, E., Popp, T.: Power analysis attacks: Revealing the secrets of smart cards, vol. 31. Springer Science & Business Media (2008)
- [37] Massey, J.L.: Minimal codewords and secret sharing. In: Proceedings of the 6th joint Swedish-Russian international workshop on information theory. pp. 276–279 (1993)
- [38] Massey, J.L.: Some applications of coding theory in cryptography. Codes and Ciphers: Cryptography and Coding IV pp. 33–47 (1995)
- [39] Masure, L., Standaert, F.X.: Prouff and Rivain’s formal security proof of masking, revisited: Tight bounds in the noisy leakage model. In: Annual International Cryptology Conference. pp. 343–376. Springer (2023)
- [40] McEliece, R.J., Sarwate, D.V.: On sharing secrets and Reed-Solomon codes. Communications of the ACM **24**(9), 583–584 (1981)
- [41] Nguyen, H.H.: Physical-bit leakage resilience of linear code-based secret sharing. In: Annual International Conference on the Theory and Applications of Cryptographic Techniques. pp. 64–93. Springer (2025)
- [42] Nielsen, J.B., Simkin, M.: Lower bounds for leakage-resilient secret sharing. In: Annual International Conference on the Theory and Applications of Cryptographic Techniques. pp. 556–577. Springer (2020)
- [43] Obremski, M., Ribeiro, J., Roy, L., Standaert, F.X., Venturi, D.: Improved reductions from noisy to bounded and probing leakages via hockey-stick divergences. In: Annual International Cryptology Conference. pp. 461–491. Springer (2024)
- [44] O’Donnell, R.: Analysis of boolean functions. Cambridge University Press (2014)
- [45] Prest, T., Goudarzi, D., Martinelli, A., Passelègue, A.: Unifying leakage models on a rényi day. In: Advances in Cryptology–CRYPTO 2019: 39th Annual International Cryptology Conference, Santa Barbara, CA, USA, August 18–22, 2019, Proceedings, Part I 39. pp. 683–712. Springer (2019)
- [46] Prouff, E., Rivain, M.: Masking against side-channel attacks: A formal security proof. In: Annual International Conference on the Theory and Applications of Cryptographic Techniques. pp. 142–159. Springer (2013)
- [47] Shamir, A.: How to share a secret. Communications of the ACM **22**(11), 612–613 (1979)
- [48] Standaert, F.X., Malkin, T.G., Yung, M.: A unified framework for the analysis of side-channel key recovery attacks. In: Advances in Cryptology–EUROCRYPT 2009: 28th Annual International Conference on the Theory and Applications of Cryptographic Techniques, Cologne, Germany, April 26–30, 2009. Proceedings 28. pp. 443–461. Springer (2009)
- [49] Standaert, F.X., Pereira, O., Yu, Y.: Leakage-resilient symmetric cryptography under empirically verifiable assumptions. In: Annual Cryptology Conference. pp. 335–352. Springer (2013)
- [50] Tao, T., Vu, V.H.: Additive combinatorics, vol. 105. Cambridge University Press (2006)